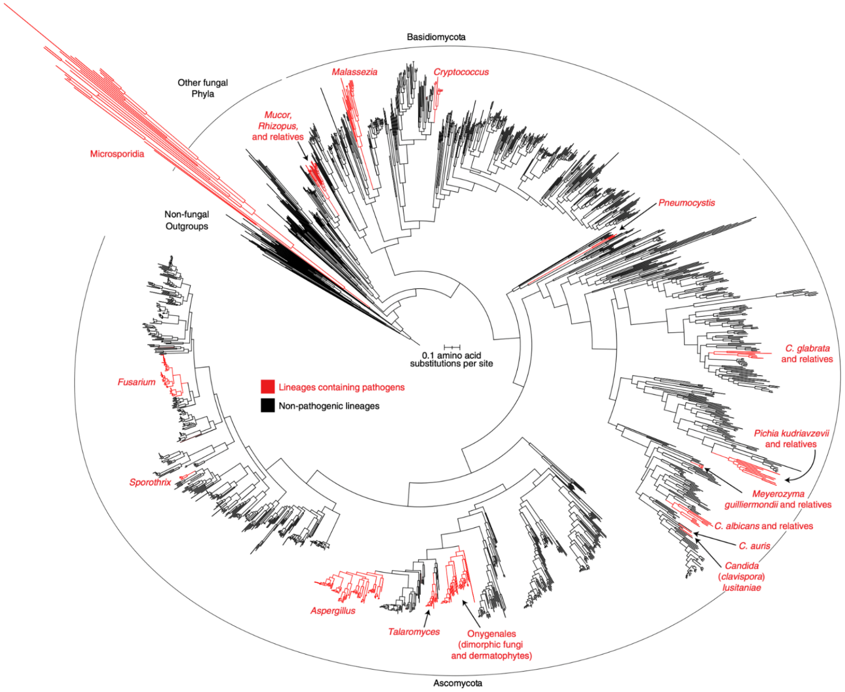
ABOUT ME

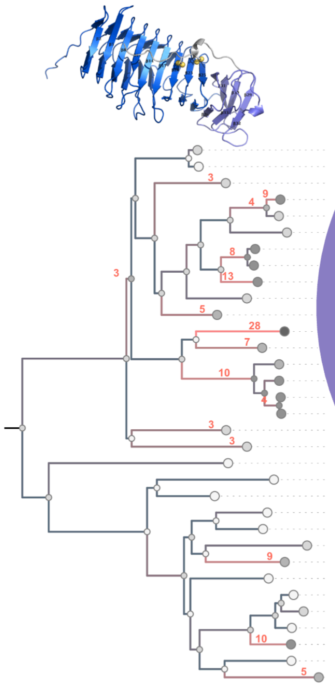
- 上海交大附中（99-02)
- 北京大学生命科学专业（02-06, 本科毕设：微流系统应用于细菌耐药性)
- 美国芝加哥大学演化与生态系（06-12, PhD thesis: evolution of gene regulatory networks and fly model of human disease）
- 美国哈佛大学系统生物学中心（12-17, Postdoc: evolution of stress response regulation in diverse yeasts）
- 美国爱荷华大学生物系（18-now, independent lab: evolution of stress response and adhesin gene families in fungal pathogens）
- https://binhe.org
- https://binhe-lab.org
- bin-he@uiowa.edu



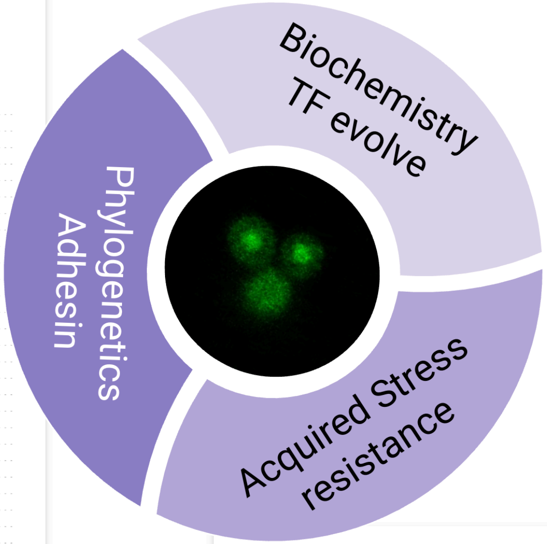
Multiple origins of pathogenic ability

- *Different mechanisms?*
- *Shared (convergent) genomic changes?*

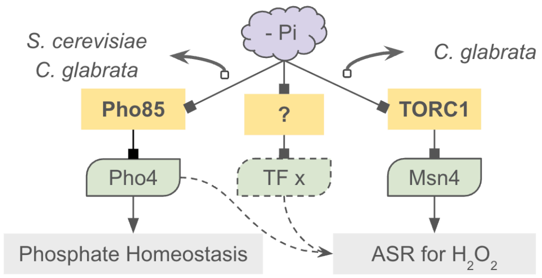
Rokas 2022 PMID: 35508719



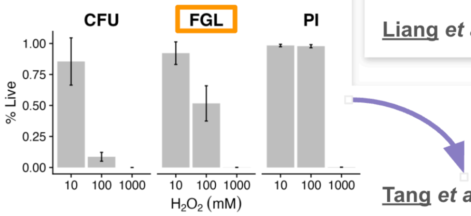
Smoak and Snyder *et al.* (2023) Genetics, PMID: 36794645



Snyder *et al.* (2024) bioRxiv, *under revision*



Liang *et al.* (2023) PLOS Pathogen, PMID: 37871123



Tang *et al.* (2024) *in preparation from MS thesis*

INTRODUCTION

WHAT IS POPULATION GENETICS? WHAT DOES IT STUDY?

POPULATION GENETICS

- A population is a group of individuals from the same species that lives in the same geographic area and that actually or potentially interbreeds.

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- Allele frequency: frequency of an allele in a population.
- Population genetics studies the genetic variation in populations and how it changes over time

WHY STUDY POPULATION GENETICS?

Biological evolution is the change over time in the **genetic composition of a population**, which changes due to the birth and death of individuals or the migration of individuals in or out of the population.

WHAT ARE THE POPULATION **EVENTS** DRIVING BIOLOGICAL EVOLUTION?

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- birth

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- birth
- death

WHAT ARE THE POPULATION **EVENTS** DRIVING BIOLOGICAL EVOLUTION?

- birth
- death
- migration

WHAT ARE THE FORCES DRIVING BIOLOGICAL EVOLUTION?

- mutation
- recombination
- natural selection
- migration
- (random/samplping) genetic drift

TYPES OF GENETIC VARIATION

- SNP

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- SNP
- microsatellites

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    - What are microsatellites?

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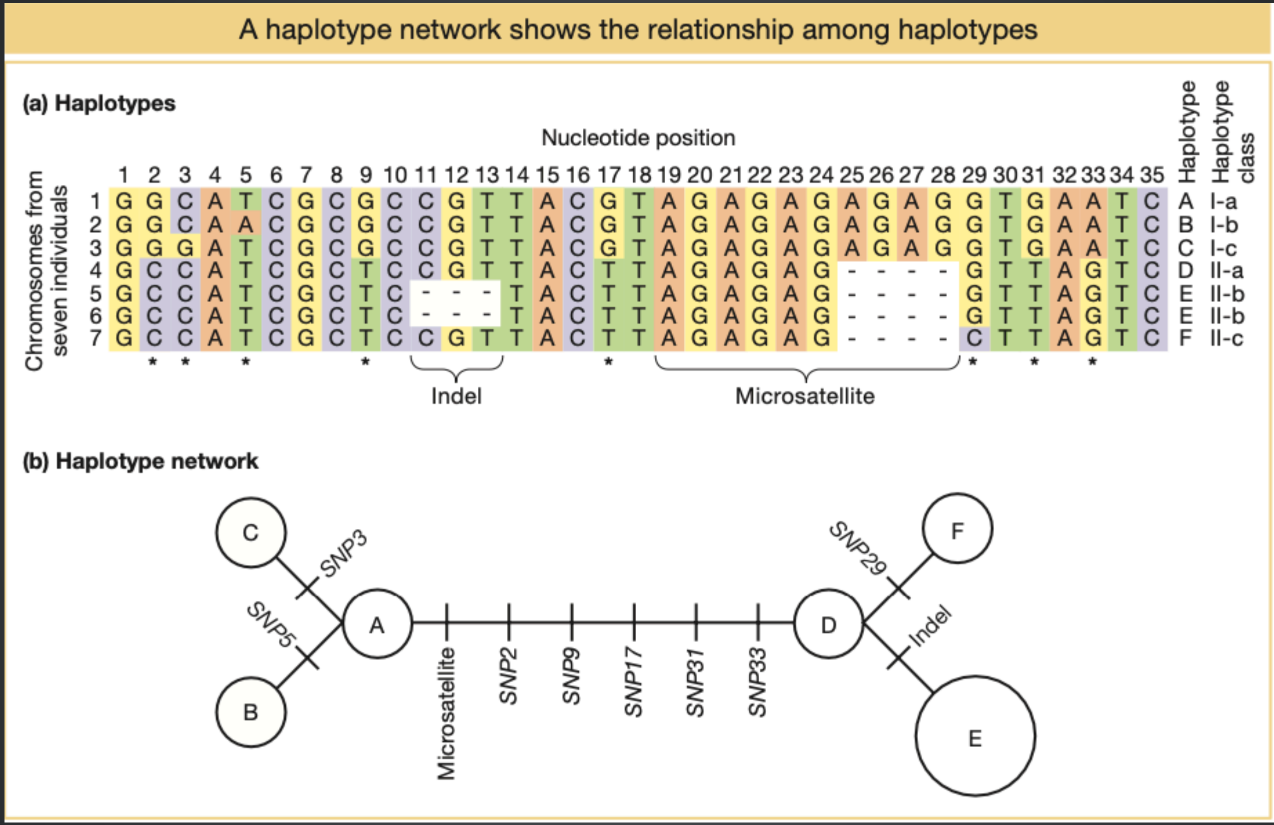
- CNV

TYPES OF GENETIC VARIATION

Haplotype

refer to the combination of alleles at multiple loci on the same chromosomal homolog.

HOW DO YOU DRAW THE NETWORK (B) BASED ON THE ALIGNMENT (A)?



1. Introduction to Genetic Analysis, ed 11, Fig. 18-4

A prevalent Y-chromosome haplotype among Asian men may trace back to Genghis Khan

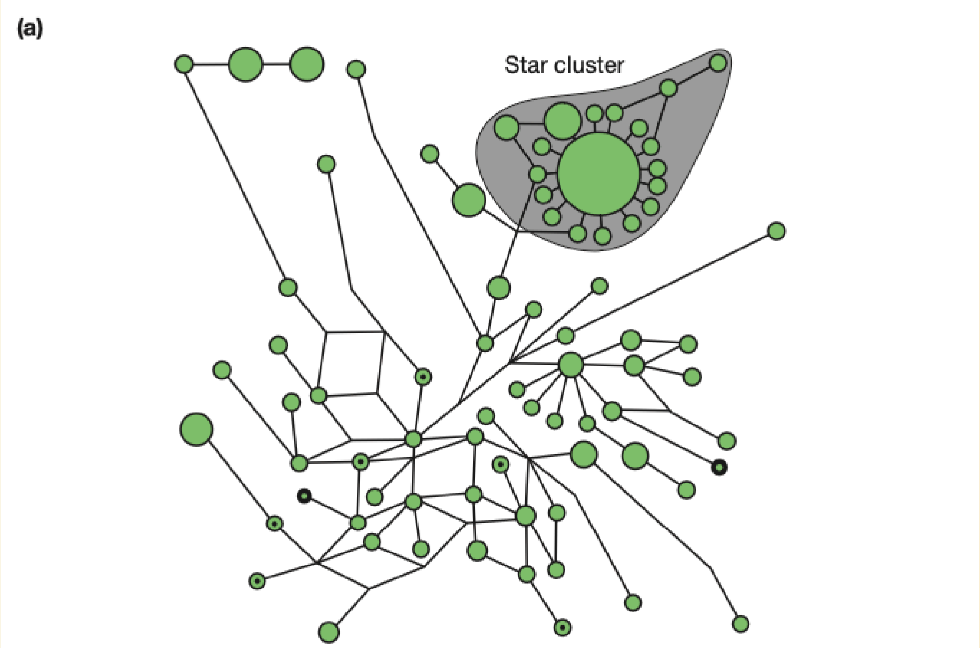
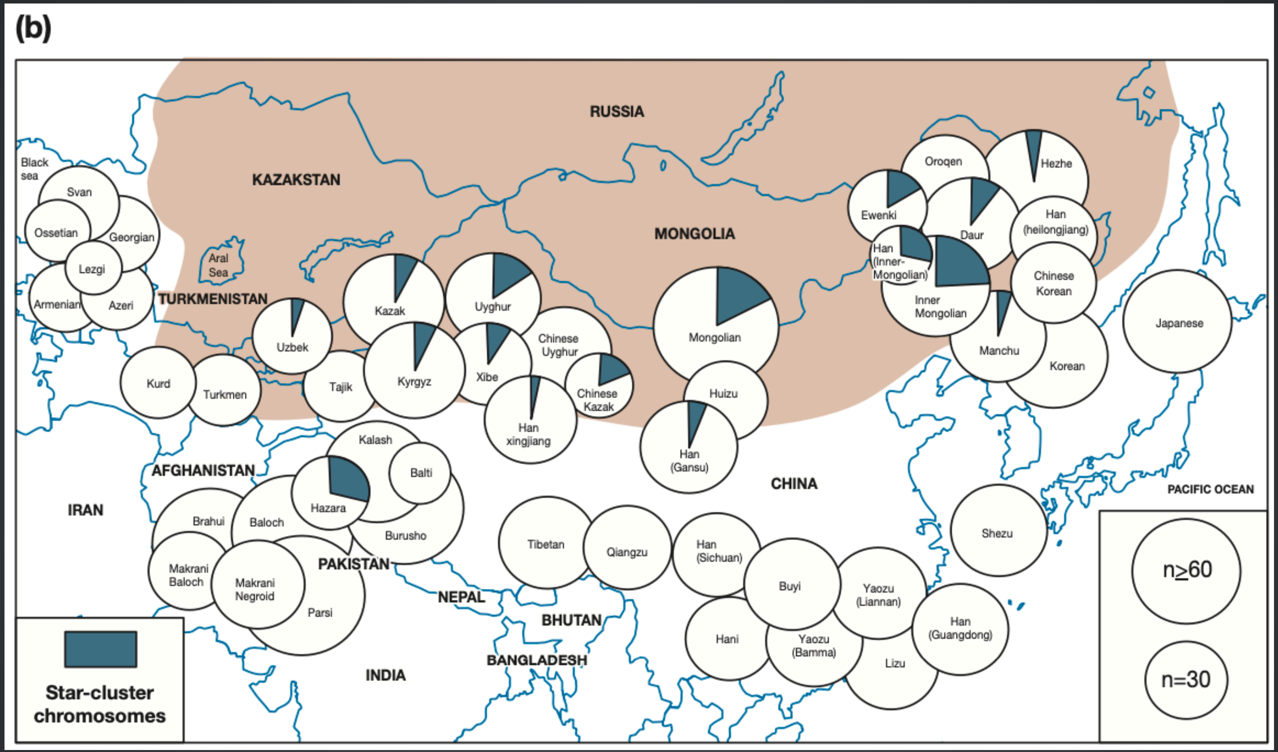


FIGURE 18-5 (a) Haplotype network for the Y chromosomes of Asian men showing the predominance of the star-cluster haplotype thought to trace back to Genghis Khan. The area of the circle is proportional to the number of individuals with the specific haplotype that the circle represents. (b) Geographical distribution of the star-cluster haplotype. Populations are shown as circles with an area proportional to sample size; the proportion of individuals in the sample carrying star-cluster chromosomes is indicated by green sectors. No star-cluster chromosomes were found in populations having no green sector in the circle. The shaded area represents the extent of Genghis Khan's empire. [*Data from T. Zerjal et al.,* Am. J. Hum. Genet. *72, 2003, 717–721.*]

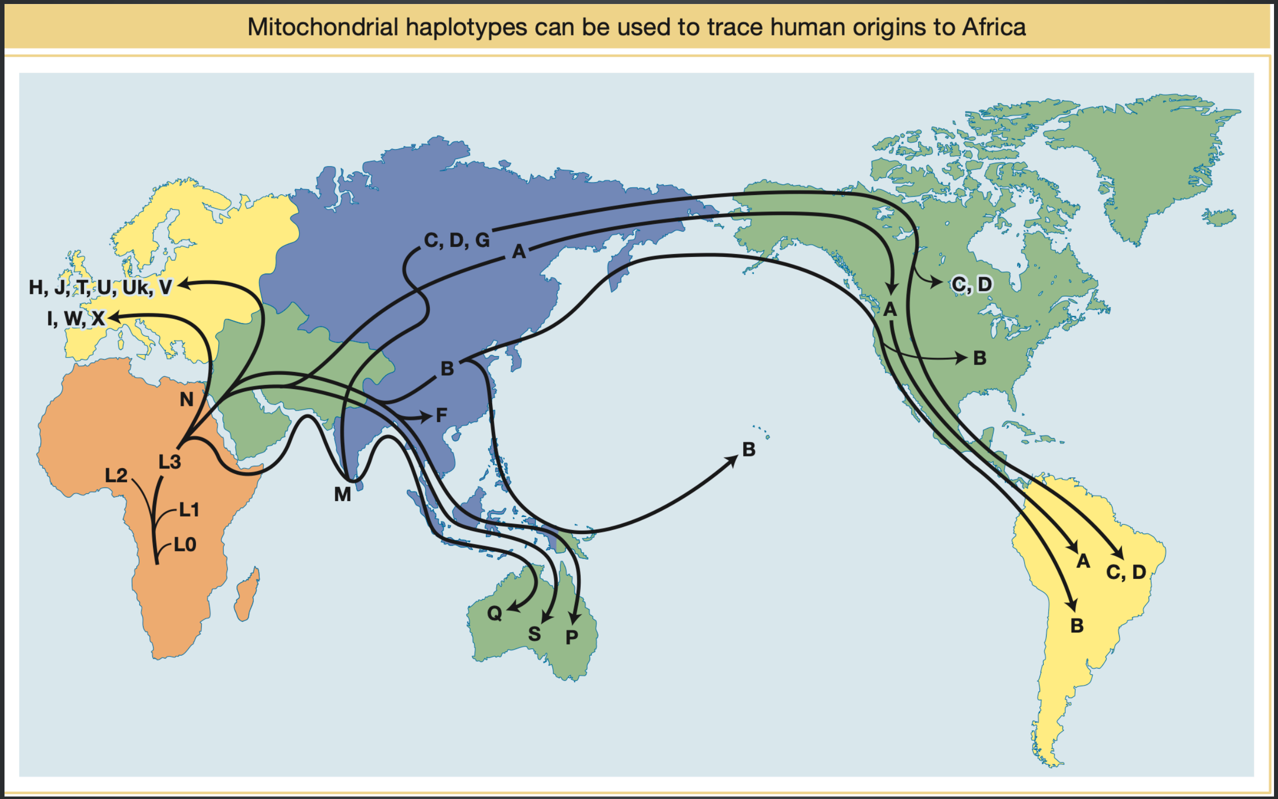
1. Introduction to Genetic Analysis, ed 11, Fig. 18-5(a)

DISTRIBUTION OF THE STAR-CLUSTER HAPLOTYPE



1. Introduction to Genetic Analysis, ed 11, Fig. 18-5(b)

WHERE DO WE COME FROM?



1: Introduction to Genetic Analysis, ed 11, Fig. 18-6

GENE POOL

- We have been evaluating genetic variations from the individual's perspective ("horizontally" as a haplotype)

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- We can also look from a site / locus / gene perspective ("vertically").
- Introduce the gene pool concept

GENE POOL

Gene pool

Sum total of all alleles in the breeding members of a population at a given time.

GENE POOL AND ALLELE FREQUENCIES

A haplotype network shows the relationship among haplotypes

(a) Haplotypes

Nucleotide position

Chromosomes from seven individuals	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Haplotype	Haplotype class
1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a
2	G	G	C	A	A	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b
3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c
4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	D	II-a
5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
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$\underbrace{11\ 12\ 13}_{\text{Indel}}$ $\underbrace{19 \ldots 28}_{\text{Microsatellite}}$

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3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c
4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	D	II-a
5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
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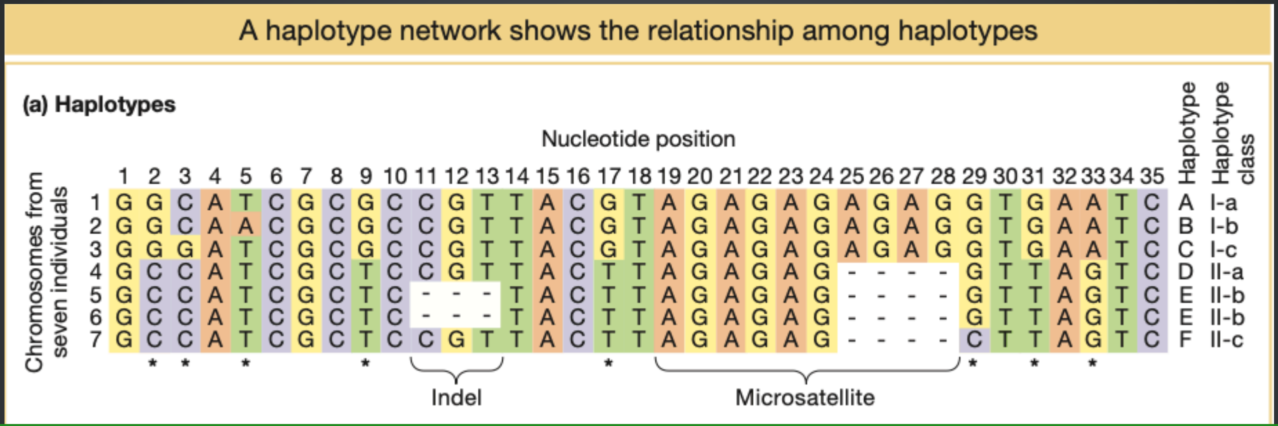
												Indel												Microsatellite													

- ==**Locus**==: "region of interest", may be an entire gene, or a single nucleotide

GENE POOL AND ALLELE FREQUENCIES



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6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
		*	*		*				*								*												*		*		*				

Indel (positions 11–13) Microsatellite (positions 19–28)

- **Locus**: "region of interest", may be an entire gene, or a single nucleotide
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5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
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- How about just looking at site 9? What's the frequency of G?

GENE POOL AND GENOTYPE FREQUENCIES

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    - Natural selection acts on fitness <- phenotype <- genotype

GENE POOL AND GENOTYPE FREQUENCIES

The gene pool is the sum total
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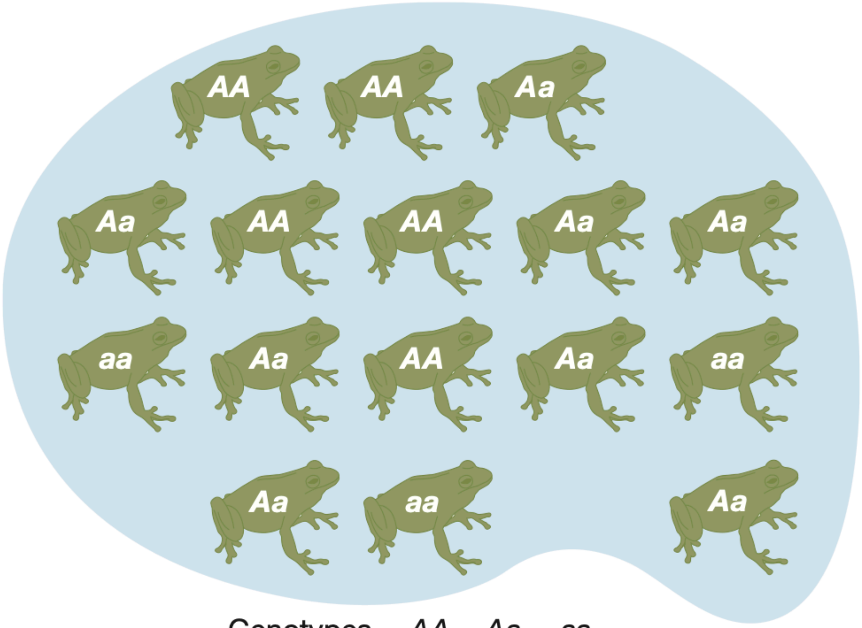
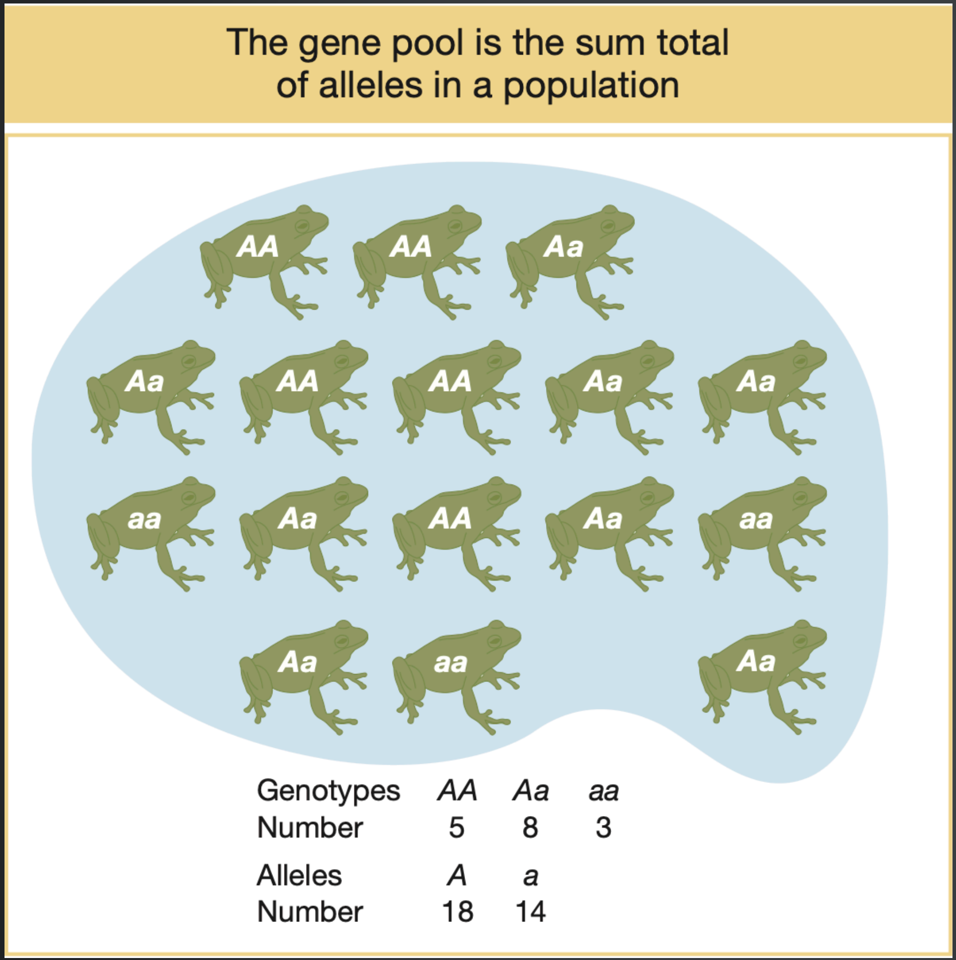


Genotypes	*AA*	*Aa*	*aa*
Number	5	8	3

Alleles	*A*	*a*
Number	18	14

GENE POOL AND GENOTYPE FREQUENCIES



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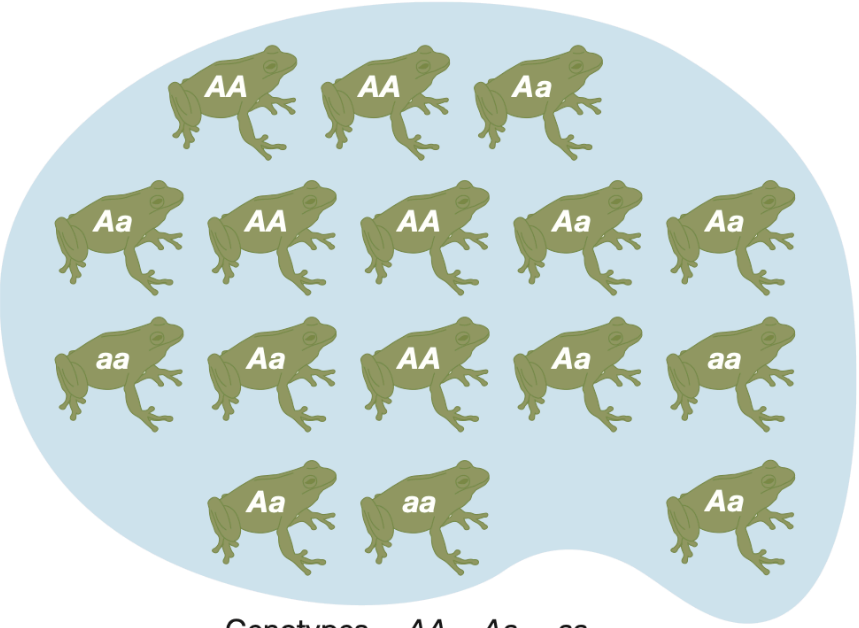

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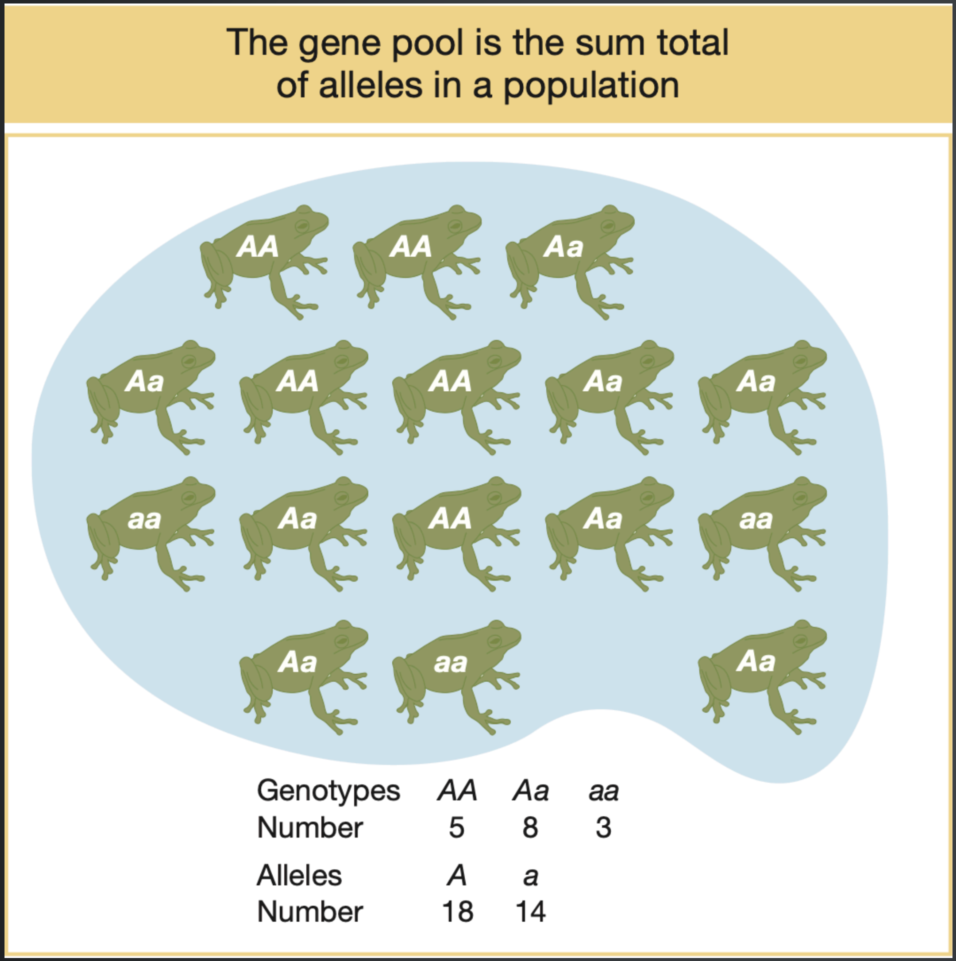

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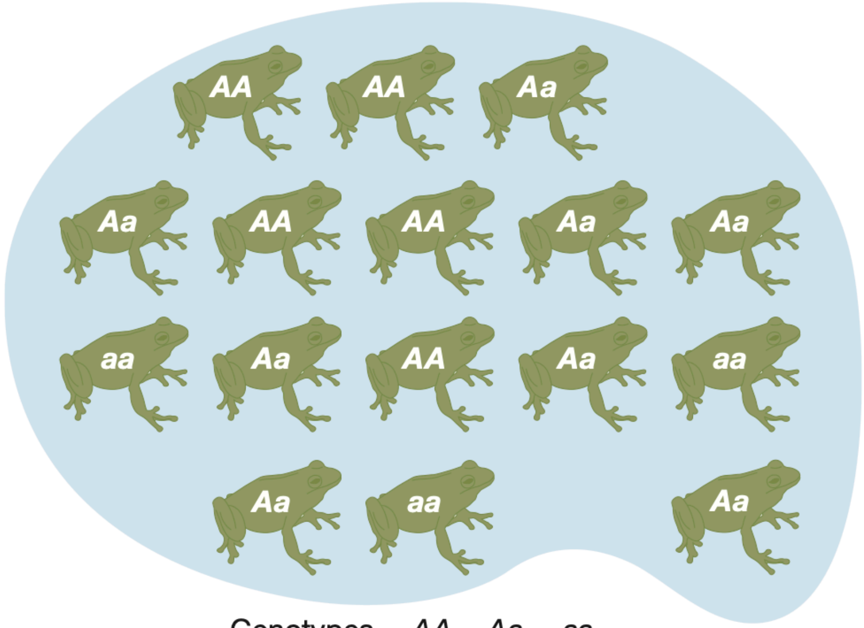

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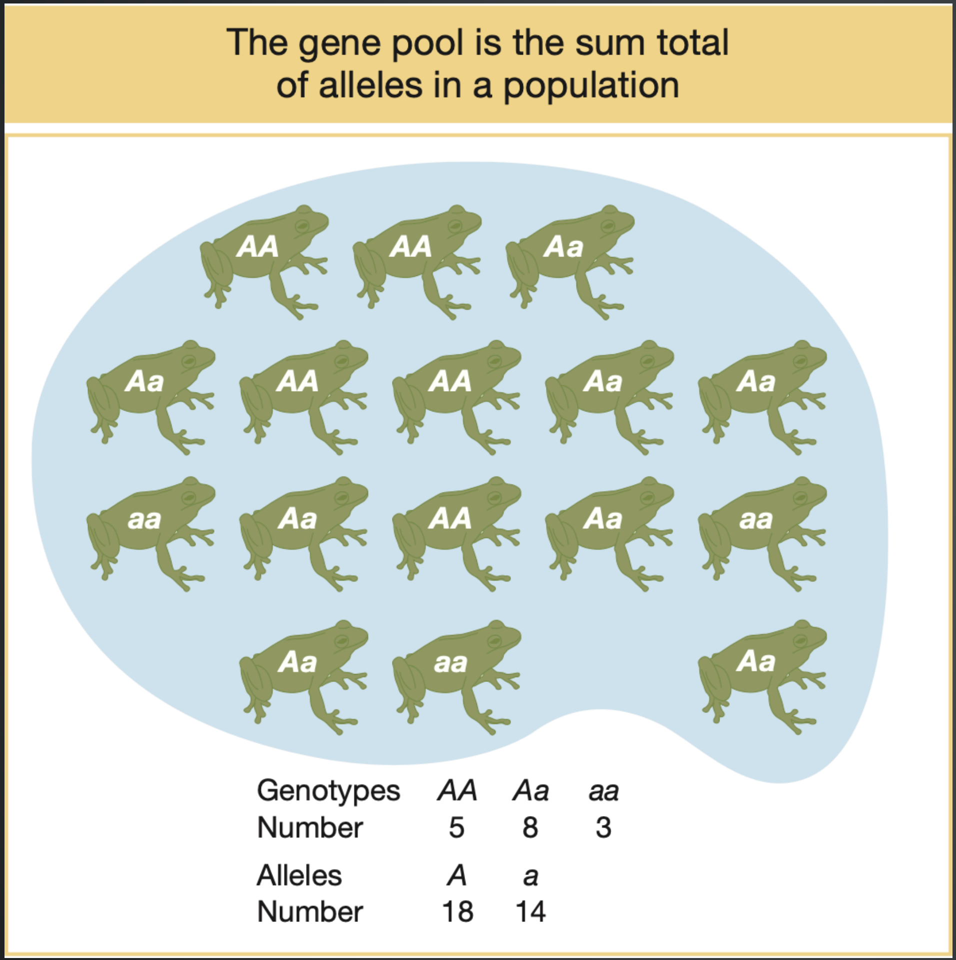
- What are the genotype frequencies?

GENE POOL AND GENOTYPE FREQUENCIES



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- What are the allele frequencies?

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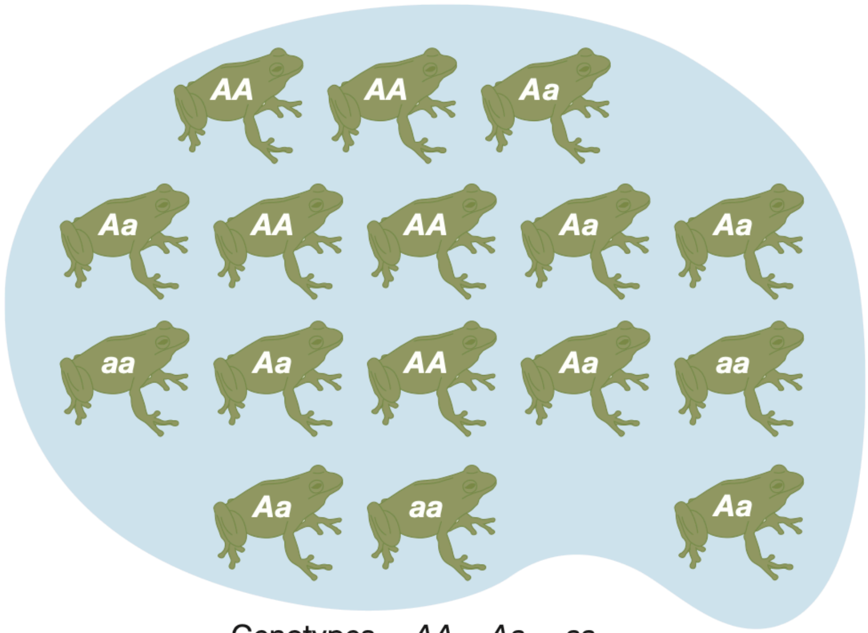

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- Knowing the allele frequency ($p_A = 18/32$), what is a "reasonable guess" of the genotype frequency (e.g., for $f_{A/A}$)?

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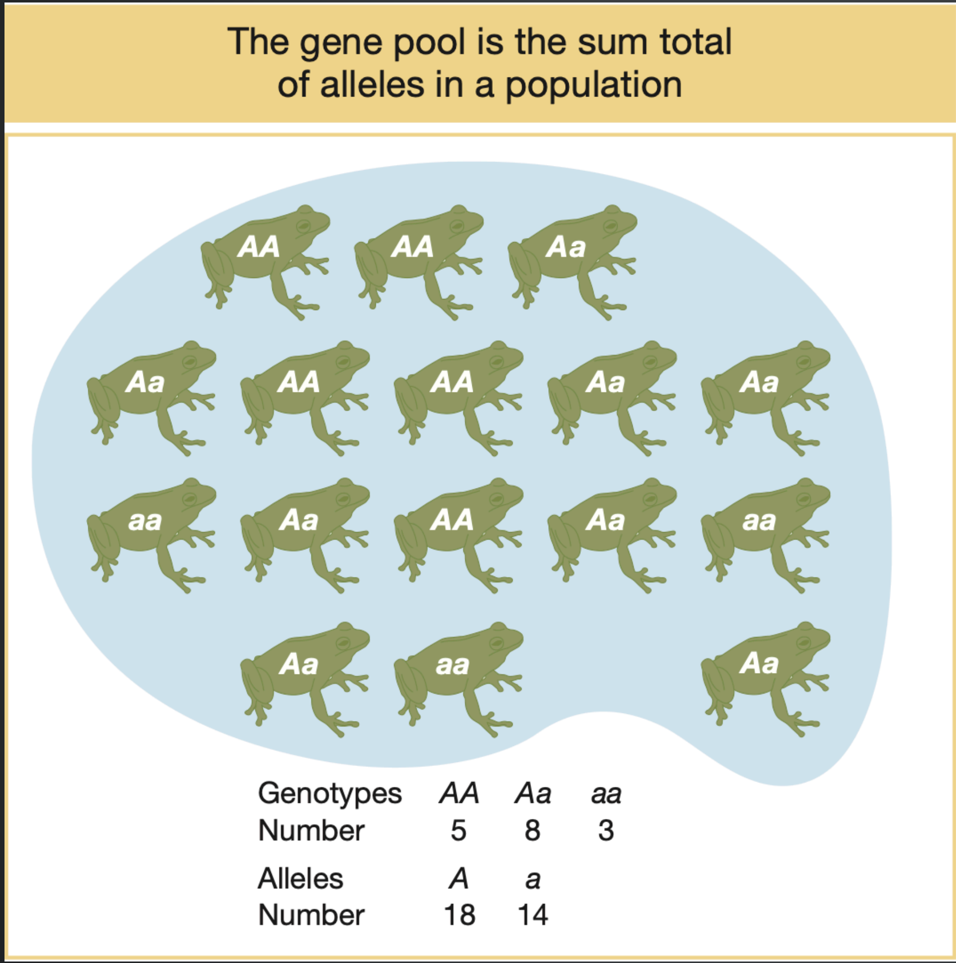

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- How close is it to the observed genotype frequency?

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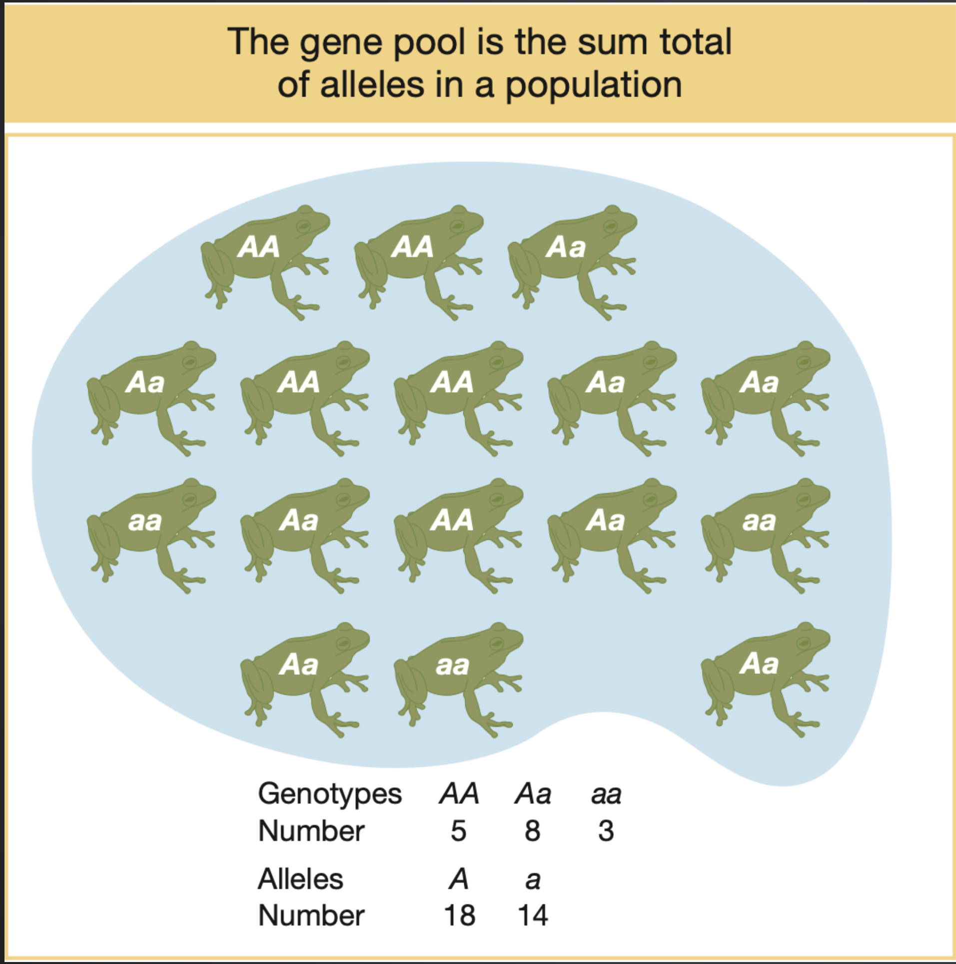

- Knowing the allele frequency ($p_A = 18/32$), what is a "reasonable guess" of the genotype frequency (e.g., for $f_{A/A}$)?
- How close is it to the observed genotype frequency?
- Now reflect on your intuitive guess, what assumptions need to be made for it to work?

HARDY-WEINBERG LAW

Suppose allele frequency of $A = p$, and frequency of $a = q$

$f_{A/A}$	$f_{A/a}$	$f_{a/a}$
p^2	$2pq$	q^2

HARDY-WEINBERG LAW

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$f_{A/A}$	$f_{A/a}$	$f_{a/a}$
p^2	$2pq$	q^2

- Assume the allele frequency of A is the same in females and males.
- The probability of a baby frog to have A/A genotype is p^2

HARDY-WEINBERG LAW

Suppose allele frequency of $A = p$, and frequency of $a = q$

$f_{A/A}$	$f_{A/a}$	$f_{a/a}$
p^2	$2pq$	q^2

- Assume the allele frequency of A is the same in females and males.
- The probability of a baby frog to have A/A genotype is p^2
- What other assumptions are needed?

HARDY-WEINBERG LAW

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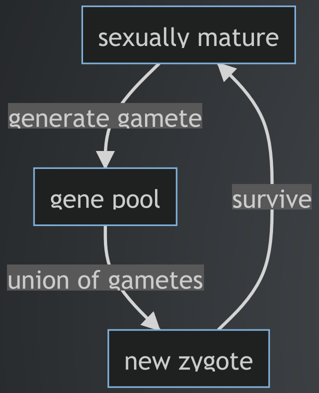
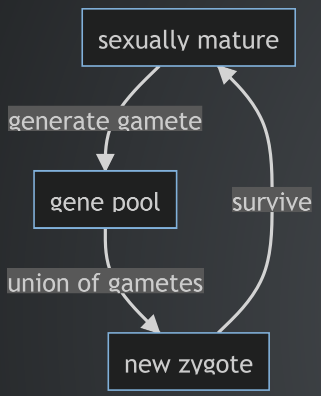
- The probability of a baby frog to have A/A genotype is p^2
- What other assumptions are needed?
- Will any evolutionary force, such as migration and natural selection, affect the above expectation?

HARDY-WEINBERG LAW



HARDY-WEINBERG LAW



- The prediction of genotype frequency from the allele frequency only relies on the step of forming new zygotes.

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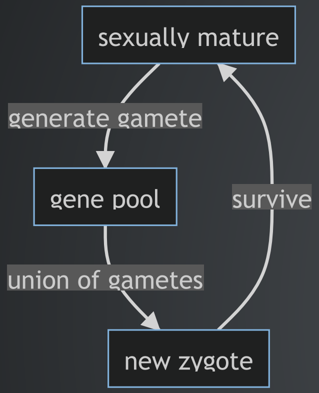

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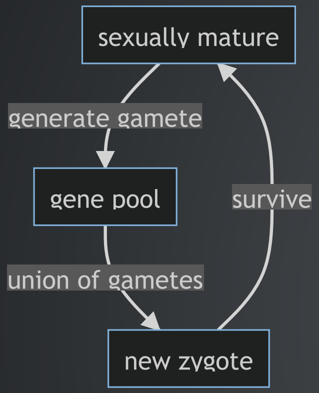

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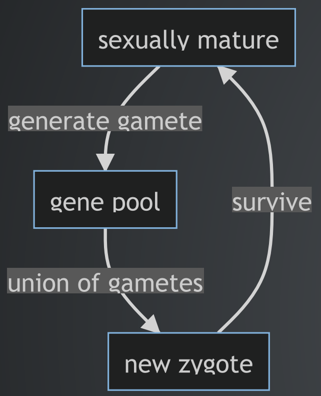

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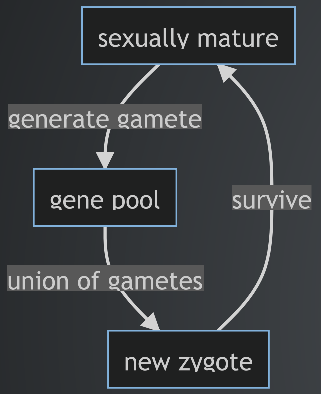

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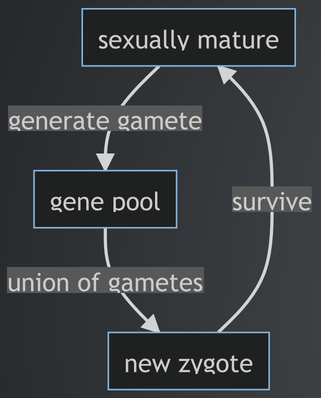

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- For the prediction to hold, we just need:
    - random mating **with respect to the locus of interest**
    - same allele frequencies in both sexes
- Under these assumptions, the expected genotypic frequency distribution is achieved *instantaneously* upon mating
- Often hear about "no selection, no migration" only needed for HW frequencies to represent a **long term equilibrium**, known as Hardy Weinberg Equilibrium

CONSEQUENCES OF HARDY WEINBERG LAW

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CONSEQUENCES OF HARDY WEINBERG LAW

1. Genotype frequencies can be estimated easily from allele frequencies.
2. Genetic variation can be maintained in a population.
3. There will be NO EVOLUTION when all of the following conditions are met

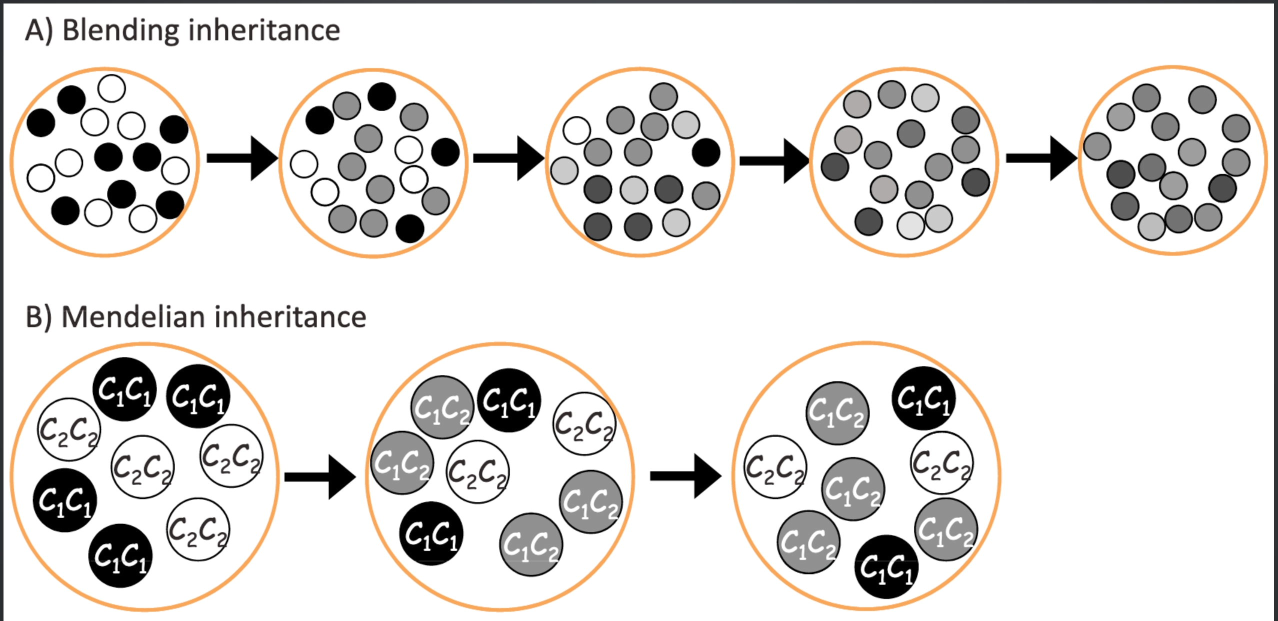
   - Infinite population size (no drift); no mutation; no selection; no migration; random mating; equal allele frequencies in males and females
   - when all of the above are true, allele frequencies do not change from generation to generation

HISTORY OF HARDY-WEINBERG LAW

1: Masel, Joanna. 2012. "Rethinking Hardy–Weinberg and Genetic Drift in Undergraduate Biology." *BioEssays* 34 (8): 701–10. https://doi.org/10.1002/bies.201100178.

HISTORY OF HARDY-WEINBERG LAW

- Maybe the original Weinberg paper?



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USING HW LAW TO ESTIMATE HETEROZYGOTE FREQUENCY

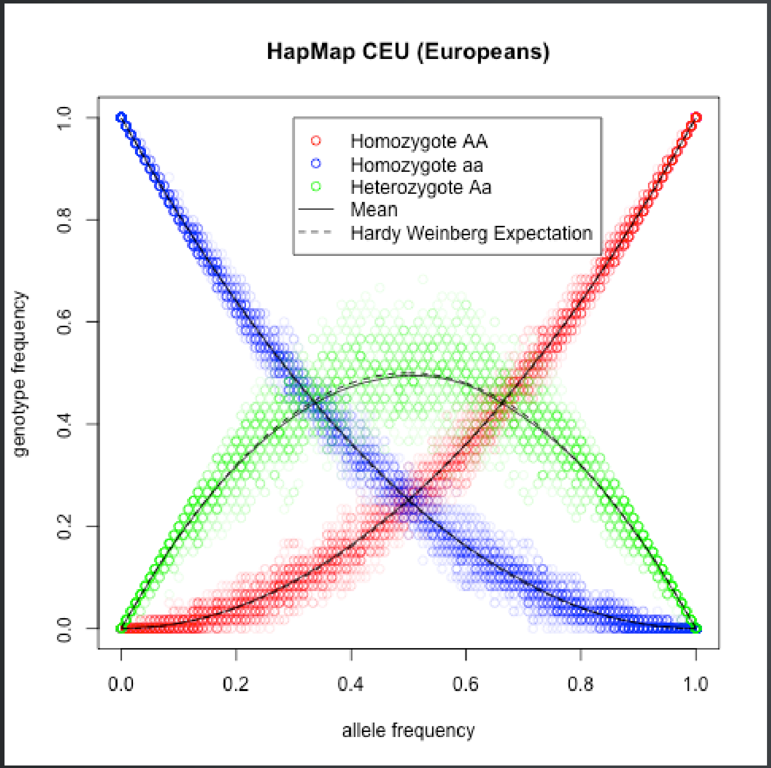
Example: cystic fibrosis, an autosomal recessive disease, has an incidence of 1 in 2500 in northern Europeans. Estimate the frequency of carriers.

$$q^2 = 1/2500,$$

So, $q = 1/50 = 0.02$, $p = 1 - q = 0.98$

$2pq = 2 \times 0.98 \times 0.02 = 0.0392$, or 1 in 25 people.

HARDY-WEINBERG EQUILIBRIUM



- Solid line: mean genotype frequency; dashed line: HWE prediction. Source
- While HWE sounds too ideal to be a good fit to real data, researchers found that it actually holds remarkably well within populations. Since the main assumption for HWE to be true is that mating is random, how should one understand this?

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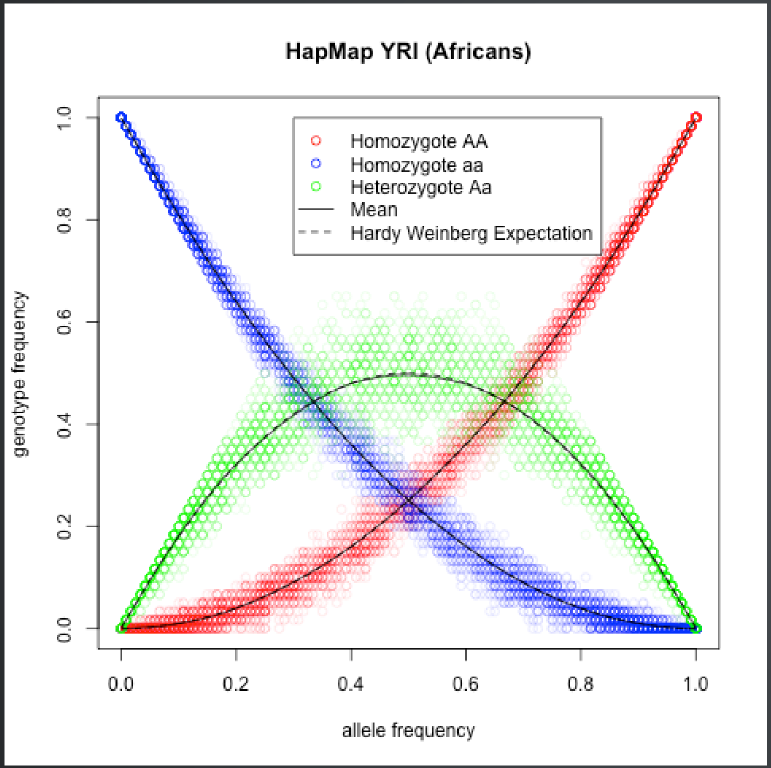

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HWE EXERCISE

In a human population, the genotype frequencies at one locus are 0.5 AA, 0.4 Aa, and 0.1 aa. The frequency of the A allele is:

A) 0.20.

B) 0.32.

C) 0.50.

D) 0.70.

E) 0.90.

HWE EXERCISE

Question

On the coastal islands of British Columbia there is a subspecies of black bear (*Ursus americanus kermodei*, Kermode's bear). Many members of this black bear subspecies are white; they're sometimes called spirit bears. These bears aren't hybrids with polar bears, nor are they albinos. They are homozygotes for a recessive change at the MC1R gene. Individuals who are GG at this SNP are white, while AA and AG individuals are black.

The genotype counts for the MC1R polymorphism in a sample of bears from British Columbia's island populations from Ritland et al. (2001) are: $AA = 42, AG = 24, GG = 21$

What are the expected frequencies of the three genotypes under HWE?

Figure 2.5: Kermode's bear (*Ursus americanus kermodei*). It's possible that this morph is favoured as the salmon these bears eat have a harder time seeing the light morph (KLINKA and REIMCHEN, 2009). The adaptive value of tasting like cinnamon is unknown.

HARDY-WEINBERG LAW FOR X-LINKED LOCI

- How does HW Law apply to X-linked loci?
-
-
-
-
-

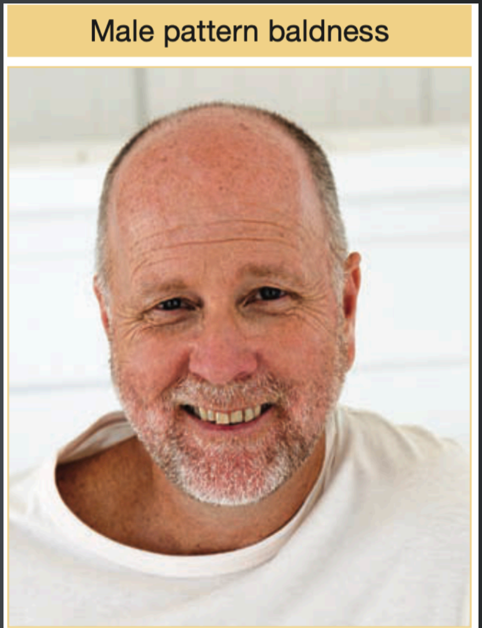

FIGURE 18-9 Individual showing male pattern baldness, an X-chromosome-linked condition. [*B2M Productions/Getty Images.*]

1: Introduction to Genetic Analysis, ed 11, Fig. 18-9

HARDY-WEINBERG LAW FOR X-LINKED LOCI

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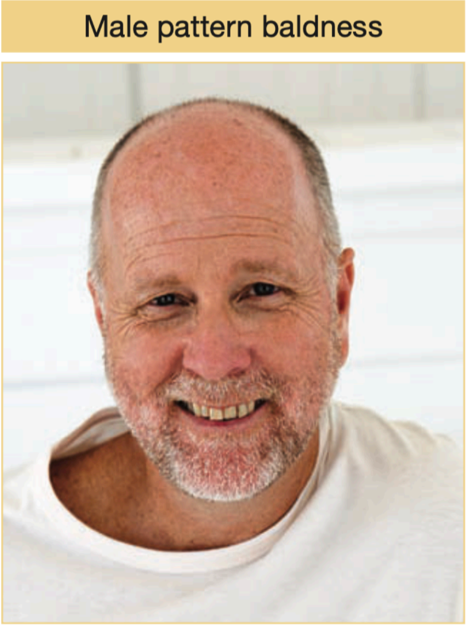

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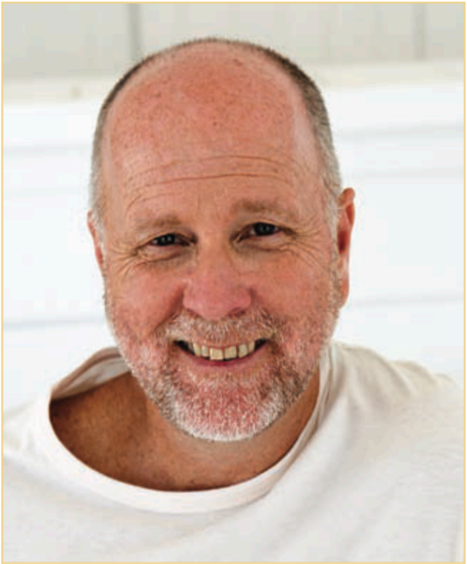
1: Introduction to Genetic Analysis, ed 11, Fig. 18-9

HARDY-WEINBERG LAW FOR X-LINKED LOCI

- How does HW Law apply to X-linked loci?
- Male-baldness is an X-linked trait.
- Androgen Receptor (AR) is an X-linked gene involved in male development.

Male pattern baldness



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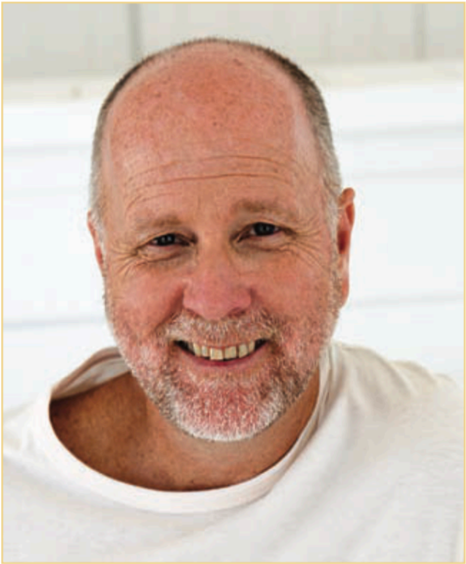
Male pattern baldness



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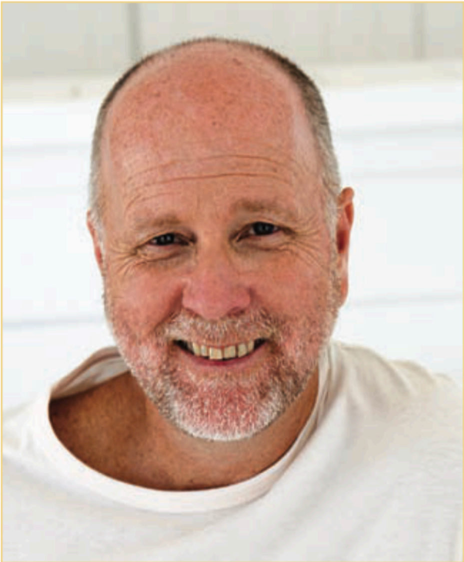

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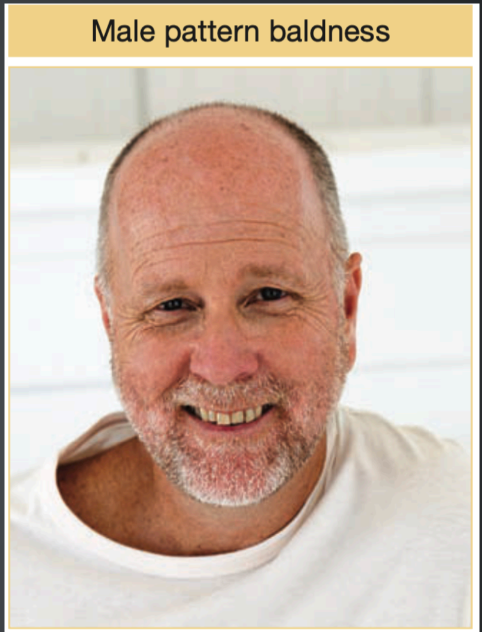
- European women?



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1: Introduction to Genetic Analysis, ed 11, Fig. 18-9

MAJOR VIOLATIONS TO HWE

ASSORTATIVE MATING

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    - Major histocompatibility complex (MHC) locus, involved in immune response to pathogens, higher heterzygosity = more diversity ≈ higher fitness.

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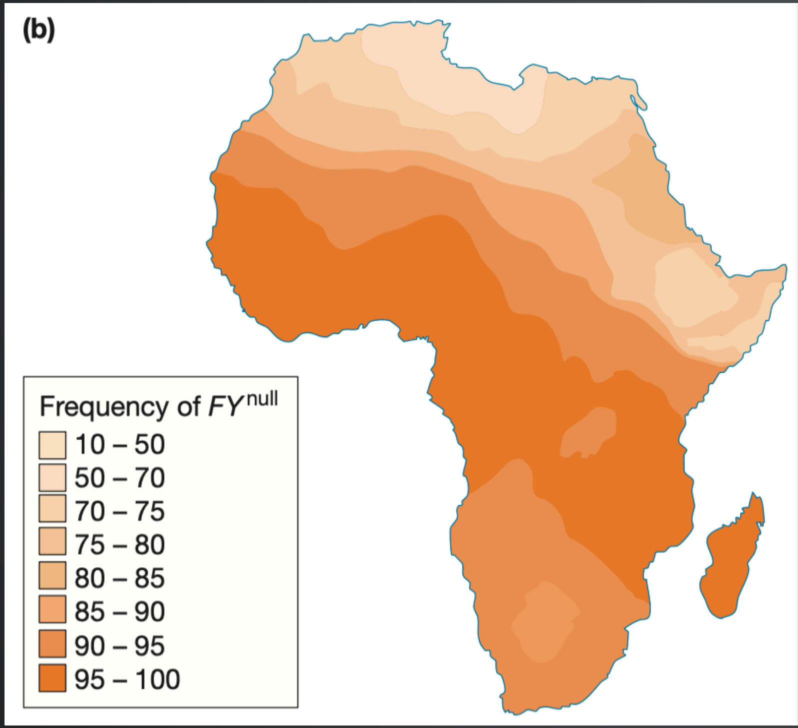
- Non-random mating ***with respect to*** the genotype at a locus
- Ex: negative or disassortative mating
    - Major histocompatibility complex (MHC) locus, involved in immune response to pathogens, higher heterzygosity = more diversity ≈ higher fitness.
    - "sweaty shirt test": MHC alleles affect mating choice, prefer individuals with different MHC alleles

MAJOR VIOLATIONS TO HWE

ISOLATION BY DISTANCE



Frequency variation for the FYnull allele of the Duffy blood group locus in Africa. [Data from P. C. Sabeti et al., Science 312, 2006, 1614–1620.]

question

what's the consequence of isolation by distance?

MAJOR VIOLATIONS TO HWE

INBREEDING

MAJOR VIOLATIONS TO HWE

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- Mating between *close* relatives.

MAJOR VIOLATIONS TO HWE

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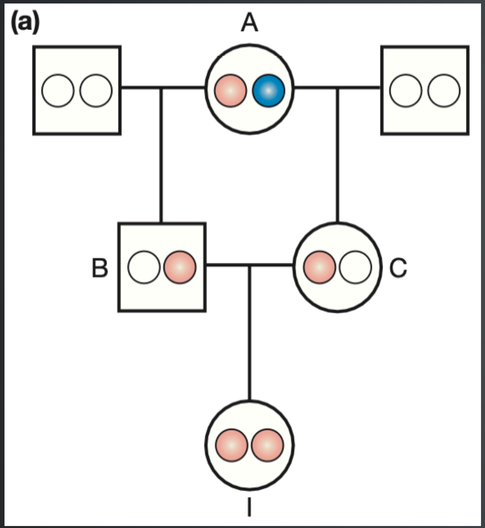
MEASURING INBREEDING



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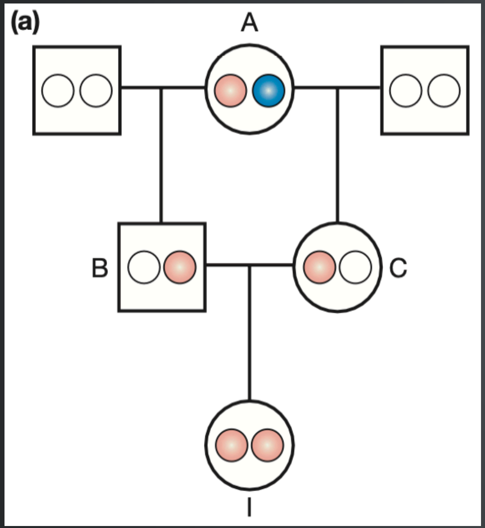

- **Inbreeding coefficient** (F): probability that two alleles in an individual trace back to the same copy in a common ancestor.

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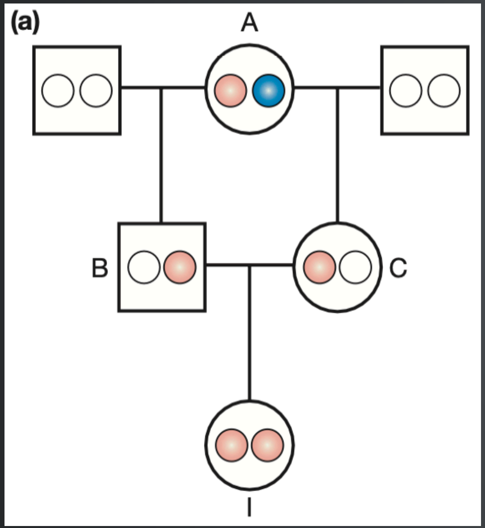

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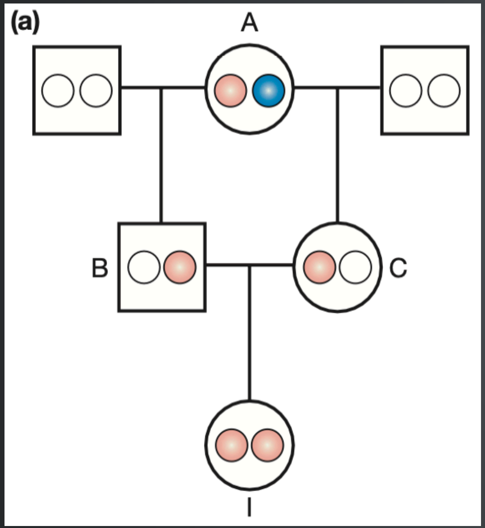

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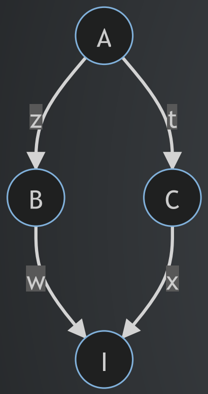
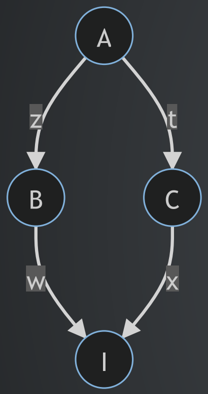
- **Identical-by-Descent** (IBD): define two alleles to be IBD if they are identical due to transmission from a common ancestor in the past few generations
- A "closed loop" indicates inbreeding.

MEASURING INBREEDING



- What's the probability that individual I is inbred? (calculating its inbreeding coefficient)

MEASURING INBREEDING



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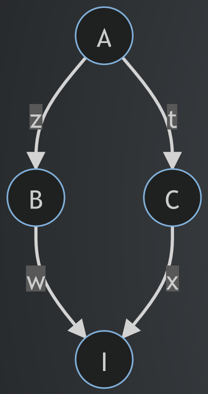

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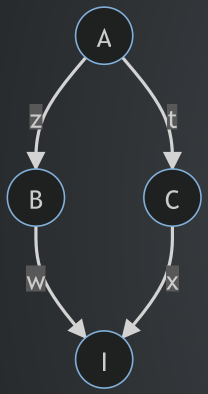

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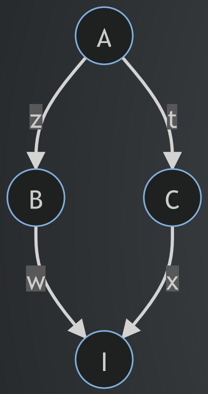

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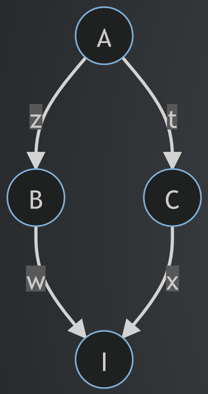

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- $P(x \sim t) = \frac{1}{2}; P(w \sim z) = \frac{1}{2}$
- $P(z \sim t) = \frac{1}{2} + \frac{1}{2}F_A$
- $F_I = P(z \sim t) \times P(w \sim z) \times P(x \sim t) = (\frac{1}{2})^3(1 + F_A)$

CONSEQUENCES OF INBREEDING

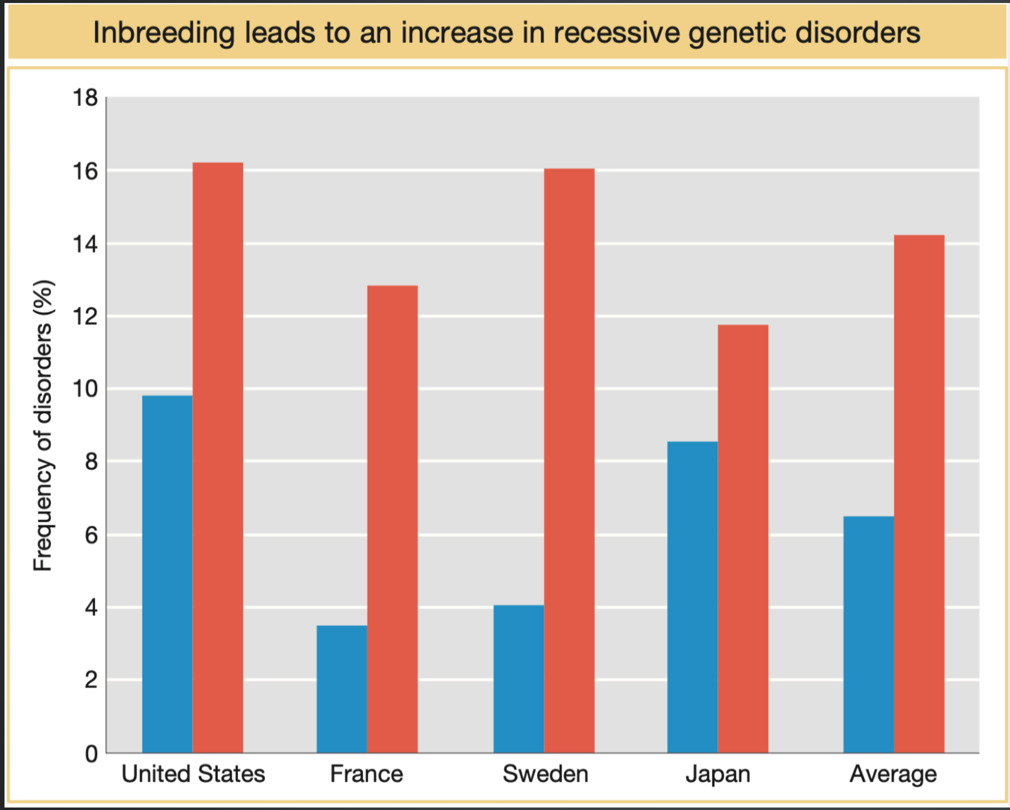
Examples of inbreeding effect on frequency of homozygous recessives

TABLE 18-3 Number of Homozygous Recessives per 10,000 Individuals for Different Allele Frequencies (q)

Mating	F	$q = 0.01$	$q = 0.005$	$q = 0.001$
Unrelated parents	0.0	1.00	0.25	0.01
Parent–offspring or brother–sister	1/4	25.75	12.69	2.51
Half-sib	1/8	13.38	6.47	1.26
First cousin	1/16	7.19	3.36	0.63
Second cousin	1/64	2.55	1.03	0.17

1: Introduction to Genetic Analysis, ed 11, Table 18-3

CONSEQUENCE OF INBREEDING



Frequency of genetic disorders among children of unrelated parents (blue columns) compared to that of children of parents who are first cousins (red columns).

Data from C. Stern, Principles of Human Genetics, W. H. Freeman, 1973.

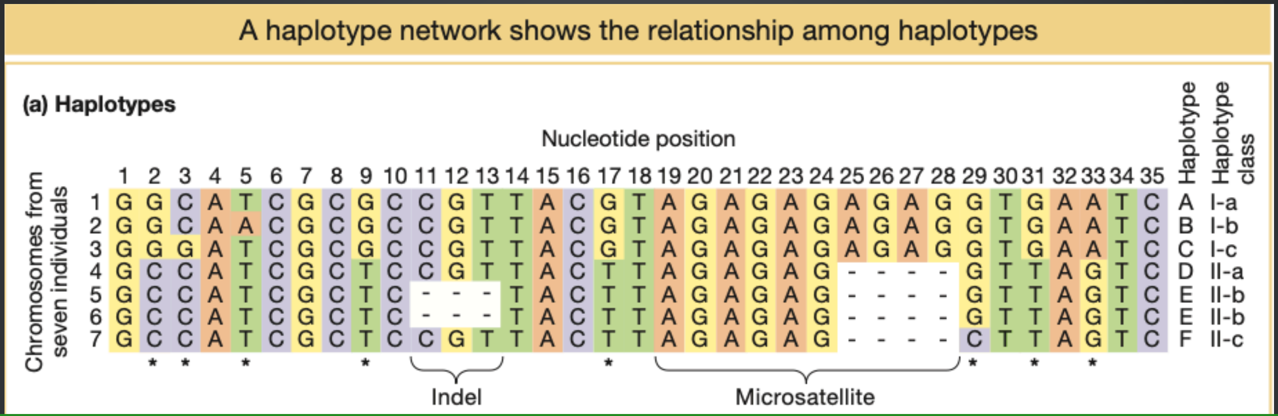
1: Introduction to Genetic Analysis, ed 11, Fig. 18-13

REVIEW OF MON LECTURE

Question

1. Describe Hardy-Weinberg Equilibrium according to your understanding. What does it concern and what are its key assumptions?
2. Describe two ways for measuring the level of genetic variation.
3. Draw a cartoon haplotype network to describe the two hypotheses in the journal club paper (*to what extent populations within continental regions exist as discrete genetic clusters versus as a genetic continuum*)

HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?



- count the number of variable sites

HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?

A haplotype network shows the relationship among haplotypes

(a) Haplotypes

Nucleotide position

Chromosomes from seven individuals	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Haplotype	Haplotype class
1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a
2	G	G	C	A	A	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b
3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c
4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	D	II-a
5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
		*	*		*				*								*												*		*		*				

$\underbrace{11\ 12\ 13}_{\text{Indel}}$ $\underbrace{19 \ldots 28}_{\text{Microsatellite}}$

- count the number of variable sites
- account for the length of the region (sequence)

HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?

A haplotype network shows the relationship among haplotypes

(a) Haplotypes

Nucleotide position

Chromosomes from seven individuals	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Haplotype	Haplotype class
1	G	G	C	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	A	I-a
2	G	G	C	A	A	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	B	I-b
3	G	G	G	A	T	C	G	C	G	C	C	G	T	T	A	C	G	T	A	G	A	G	A	G	A	G	A	G	G	T	G	A	A	T	C	C	I-c
4	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	D	II-a
5	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
6	G	C	C	A	T	C	G	C	T	C	-	-	-	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	G	T	T	A	G	T	C	E	II-b
7	G	C	C	A	T	C	G	C	T	C	C	G	T	T	A	C	T	T	A	G	A	G	A	G	-	-	-	-	C	T	T	A	G	T	C	F	II-c
		*	*		*				*								*												*		*		*				

Indel

Microsatellite

- count the number of variable sites
- account for the length of the region (sequence)
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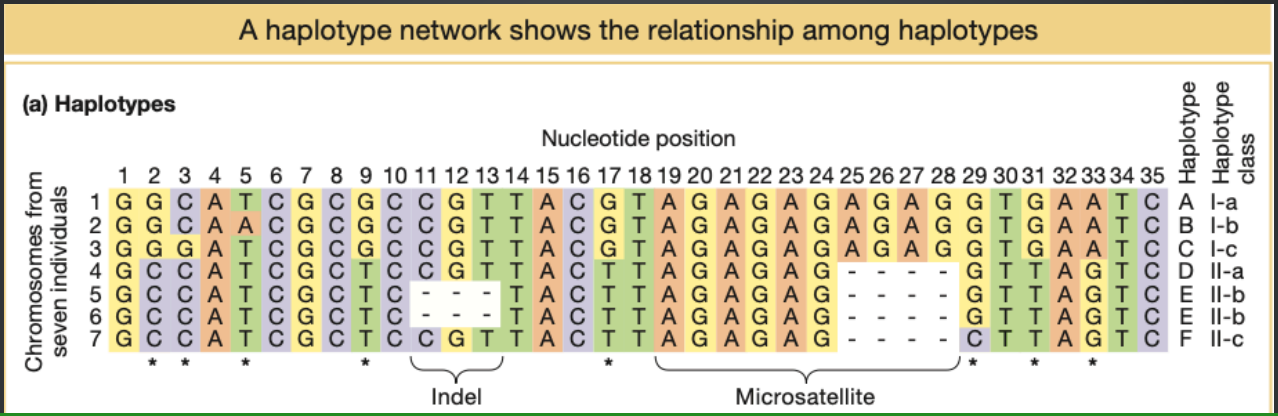
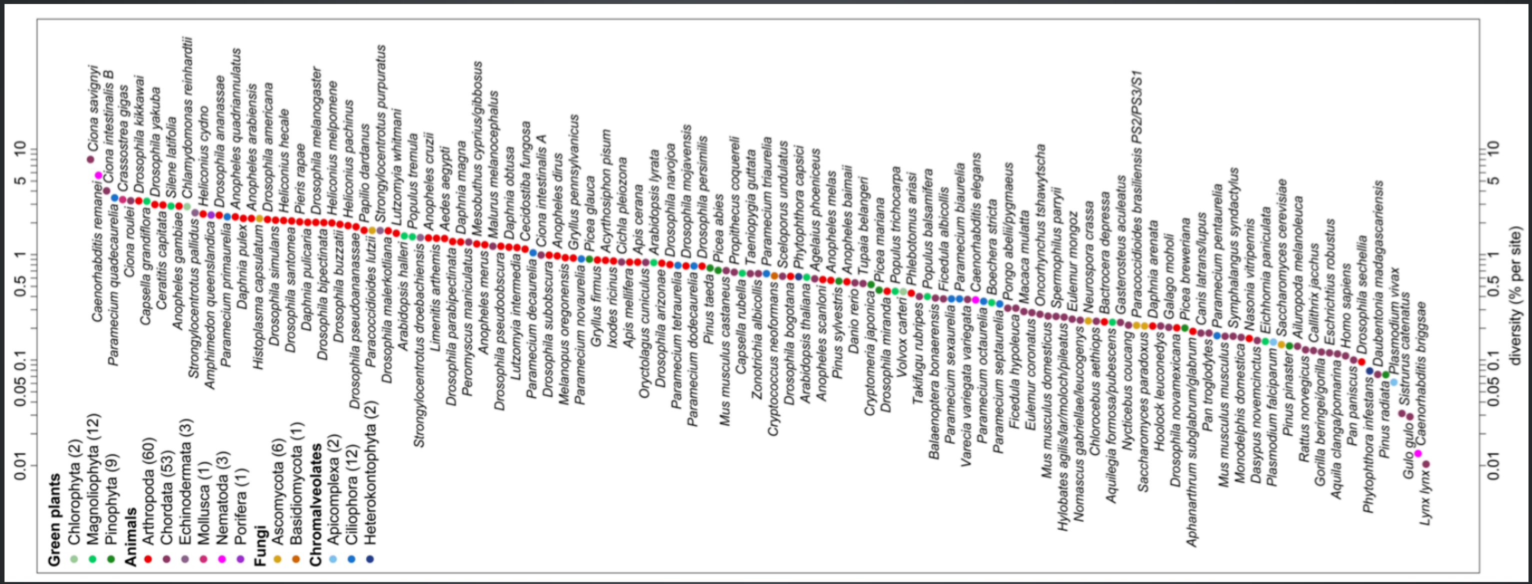
HOW DO YOU QUANTIFY THE LEVEL OF GENETIC VARIATION?



- count the number of variable sites
- account for the length of the region (sequence)
- account for the number of sequences (sample size)
- Average # of differences between two sequences

LEVEL OF GENETIC VARIATION VARY ACROSS SPECIES



1: Leffler et al. 2012. "Revisiting an Old Riddle: What Determines Genetic Diversity Levels within Species?" *PLoS Biology* 10 (9): e1001388. https://doi.org/10.1371/journal.pbio.1001388.

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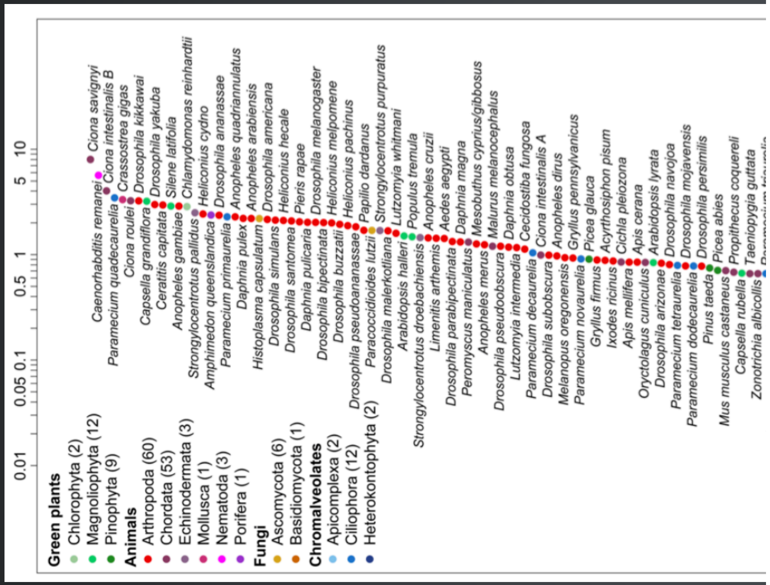

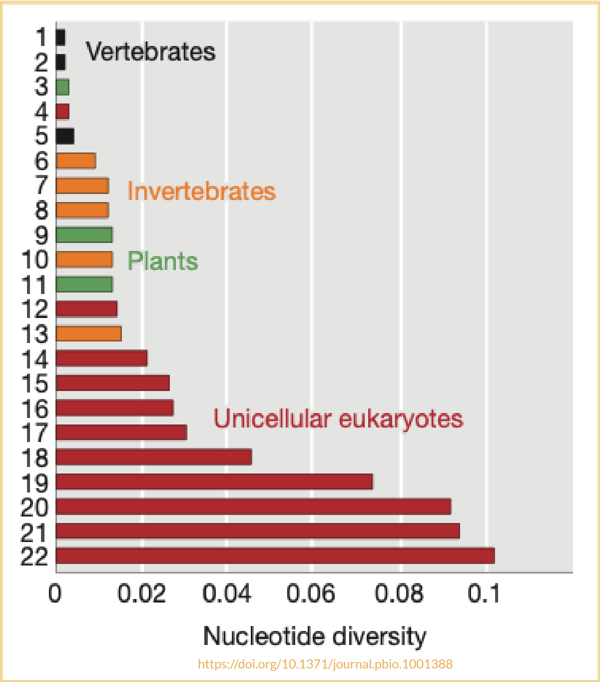

M. Lynch and J. S. Conery, Science 2003

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WHAT FORCES SHAPE THE LEVEL AND PATTERN OF GENETIC VARIATION?

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- Forces examined next

    1. mutation
    2. migration
    3. drift
    4. recombination
    5. selection

MUTATION

TABLE 18-5 Approximate Mutation Rates per Generation per Haploid Genome

Organism	SNP mutations (per bp)	Microsatellite
Arabidopsis	7×10^{-9}	9×10^{-4}
Maize	3×10^{-8}	8×10^{-4}
E. coli	5×10^{-10}	–
Yeast	5×10^{-10}	4×10^{-5}
C. elegans	3×10^{-9}	4×10^{-3}
Drosophila	4×10^{-9}	9×10^{-6}
Mouse	4×10^{-9}	3×10^{-4}
Human	3×10^{-8}	6×10^{-4}

Note: Microsatellite rate is for di- or trinucleotide repeat microsatellites.

Source: Data from multiple published studies.

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- doesn't just apply to a single nucleotide position - for any locus
- in a single generation is important - other forces can change allele forms, but they act after the formation of zygote stage
- data in human (circa 2009) gave an estimate of $\sim 3.0 \times 10^{-8}$ mutations/nucleotide/generation for a part of the Y-chromosome. if we extrapolate this to the entire human genome, we get an estimate of about 100 new mutations one would inherit from each of our parents *on average*.

MIGRATION

- In addition to mutation, migration is the other way by which new variation can be introduced into a population (not counting **new combinations**, see recombination)
- when there is isolation by distance, migration (or gene flow) is a homogenizing force, preventing allele frequencies from diverging too far

RECOMBINATION

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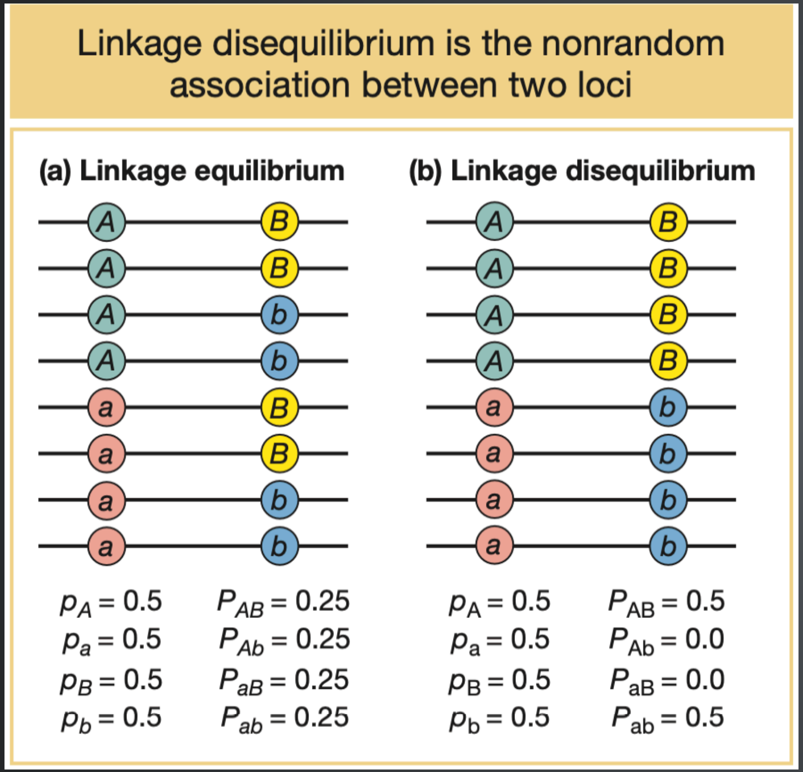
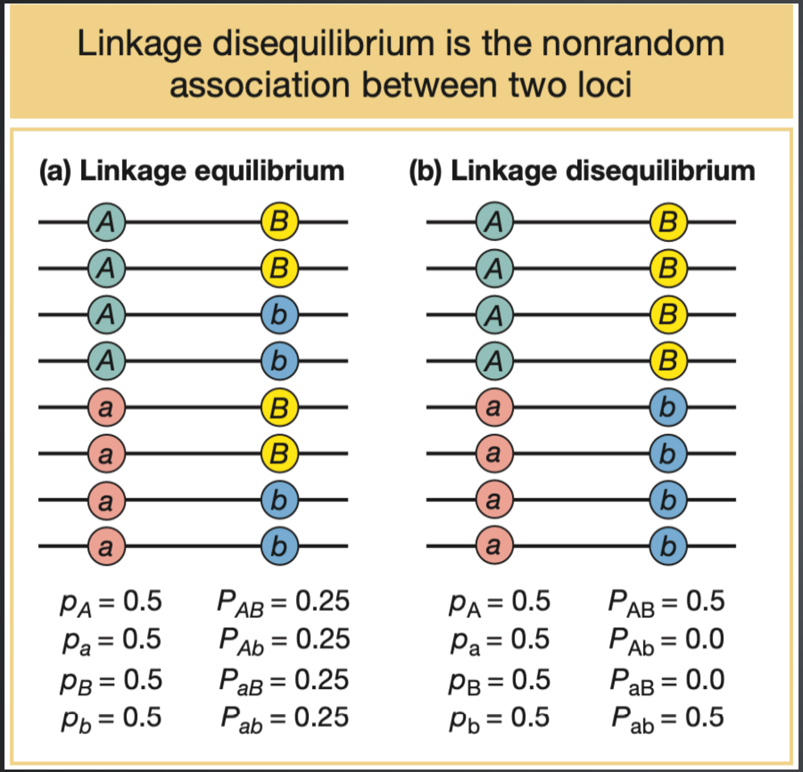
    - $P_{AB} = p_A \times p_B$ etc.
    - If the observed frequencies are very close to the expected, we say that the two loci are in **linkage equilibrium**
    - If not, then the loci are said to be in **linkage disequilibrium (LD)**.

LINKAGE DISEQUILIBRIUM



1: Introduction to Genetic Analysis, ed 11, Fig. 18-17

LINKAGE DISEQUILIBRIUM



- Define LD as the difference (D) between the observed and expected frequencies

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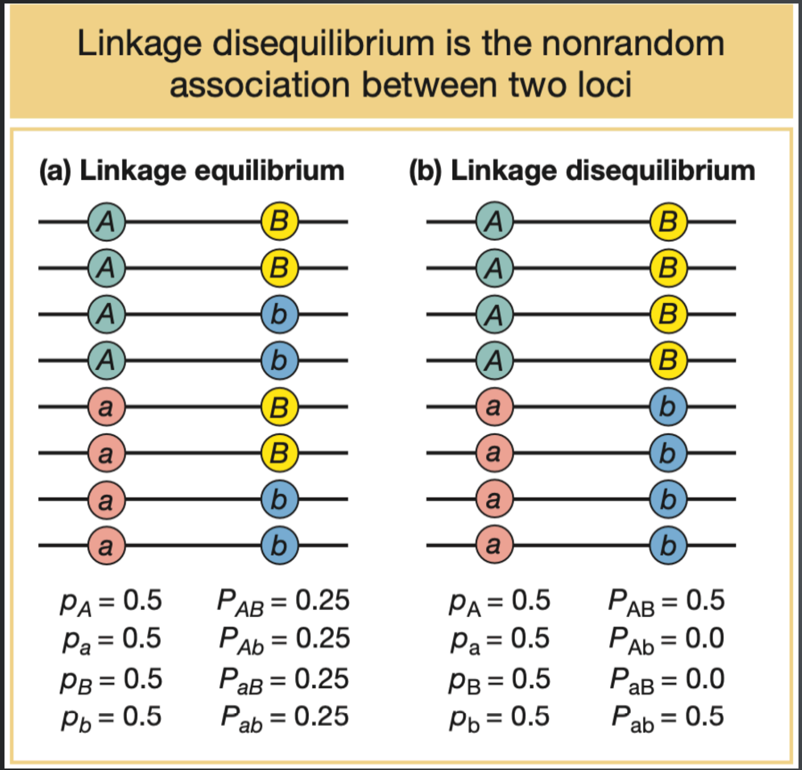
LINKAGE DISEQUILIBRIUM



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  - $D = P_{AB} - p_A p_B$

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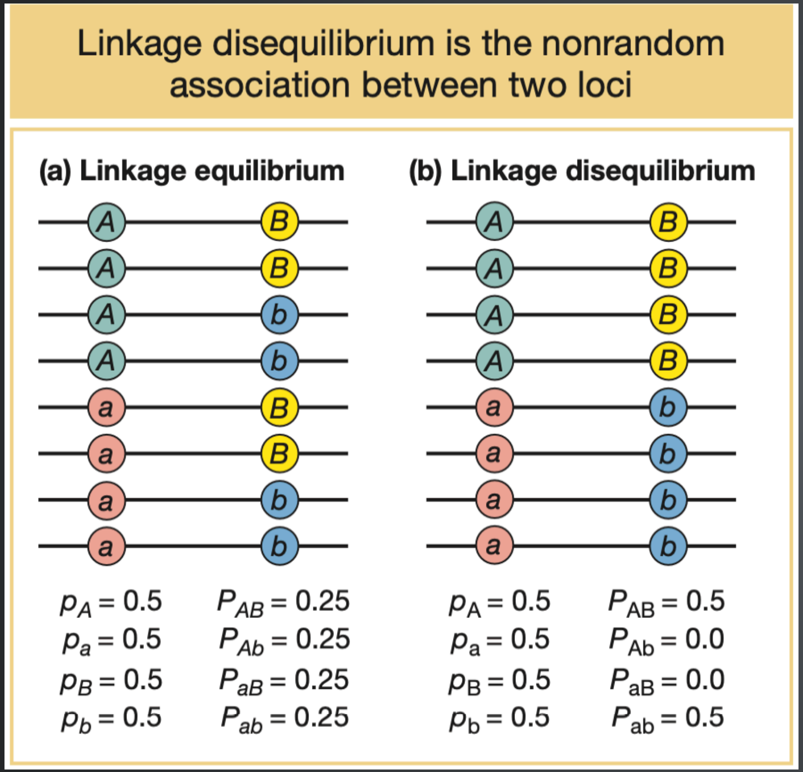
LINKAGE DISEQUILIBRIUM



- Define LD as the difference (D) between the observed and expected frequencies
  - $D = P_{AB} - p_A p_B$
- Exercise: calculate D for the two scenarios on the left

1: Introduction to Genetic Analysis, ed 11, Fig. 18-17

LINKAGE DISEQUILIBRIUM

Nucleotide variation at the *G6PD* gene in humans

Individual	Origin	Allele	SNP 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Haplotype
			A	G	A	C	C	G	C	C	C	C	C	G	G	C	T	C	A	C	
1	Southern African	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
2	Central African	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
3	Central African	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
4	African American	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
5	African American	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
6	Central African	*A-*	G	**A**	G	•	G	•	•	T	•	T	•	•	•	•	C	•	G	•	1
7	Central African	*A+*	G	•	G	•	•	•	•	•	•	T	•	•	•	•	C	•	G	•	2
8	Central African	*A+*	G	•	G	•	•	•	•	•	•	T	•	•	•	•	C	•	G	•	2
9	Central African	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	C	•	G	•	3
10	Southern African	*B*	•	•	•	•	•	•	•	•	•	•	•	A	•	•	C	T	G	•	4
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13	Southern African	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	C	T	G	•	6
14	Southern African	*B*	•	•	•	•	•	•	•	•	•	•	T	•	A	•	C	•	G	•	7
15	Central African	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	T	C	•	G	•	8
16	European	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	T	C	•	G	•	8
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18	European	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	T	C	•	G	•	8
19	Southwest Asian	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	T	C	•	G	•	8
20	East Asian	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	C	•	G	•	3
21	Native American	*B*	•	•	•	•	•	A	T	•	•	•	•	•	•	•	C	•	G	•	9
22	Southern African	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	10
23	Native American	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	10
24	Native American	*B*	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	10
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- How LD arises

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- How LD arises
  - consider a population with only AB and Ab. what happens when a new allele "a" arises at A locus, on a chromosome that already possess b. Is LD zero? or is it maximum?

LINKAGE DISEQUILIBRIUM

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  - can be used to estimate the age of mutation, how?

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  - what happens when we increase N to 2?

GENETIC DRIFT AND POPULATION SIZE

Drift

any change in allele frequencies due to sampling error, not just loss or fixation of an allele

Question

In a population with 500 individuals (N=500) and two alleles at a frequency of $p = q = 0.5$, if the next generation has 501 copies of A and 499 copies of a, **has there been genetic drift**?

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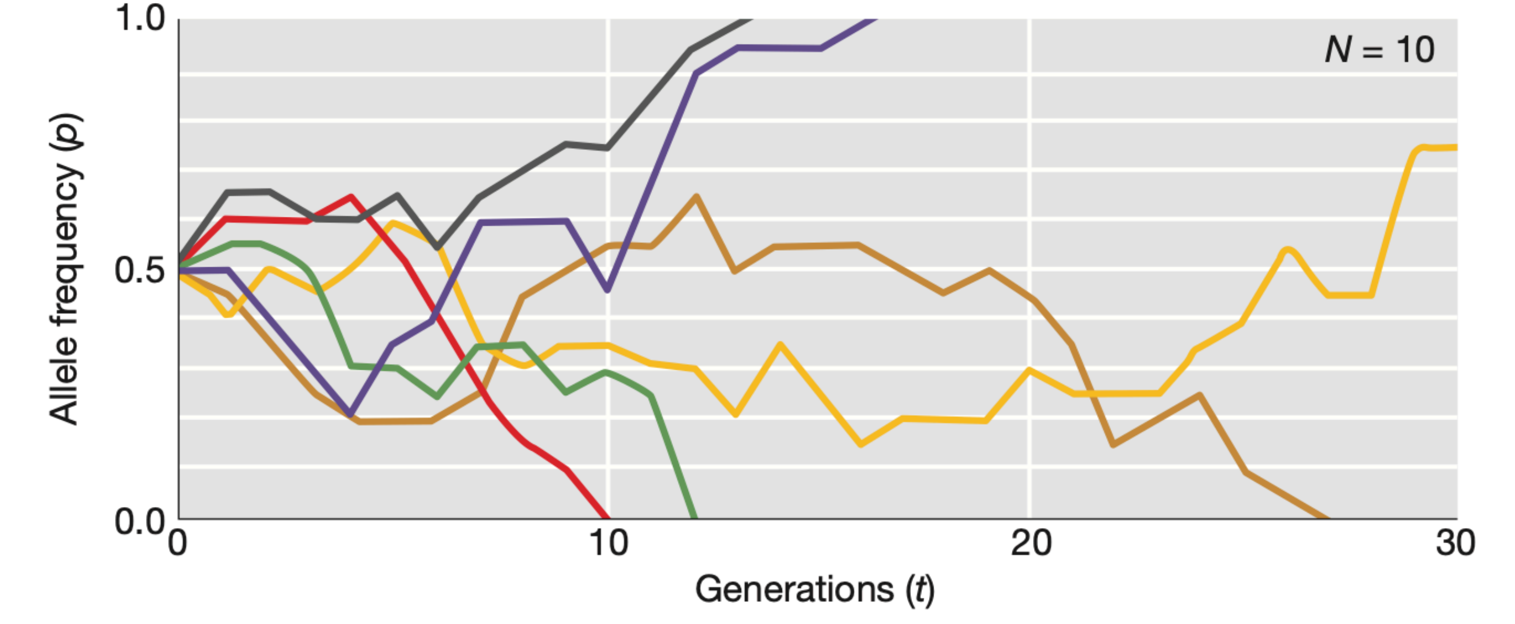
GENETIC DRIFT AND POPULATION SIZE



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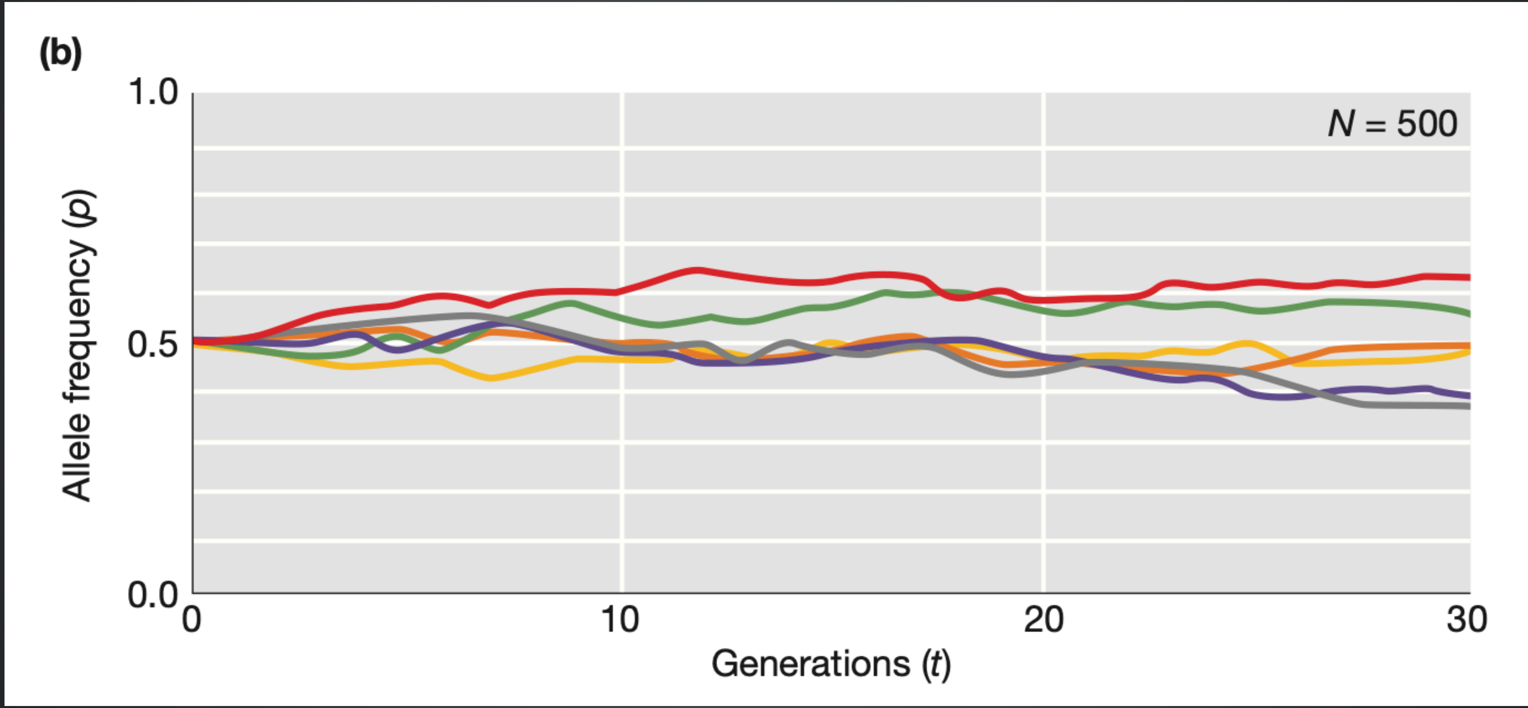

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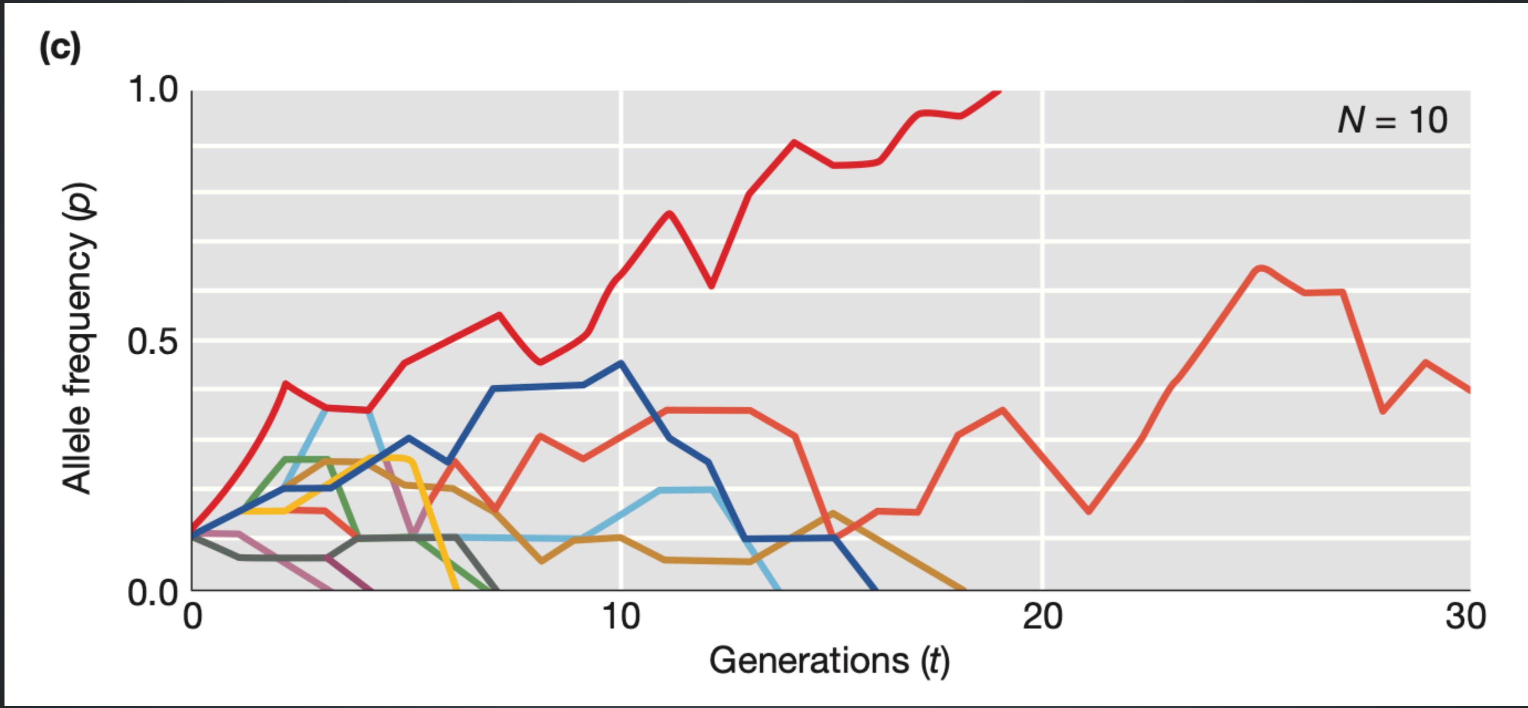

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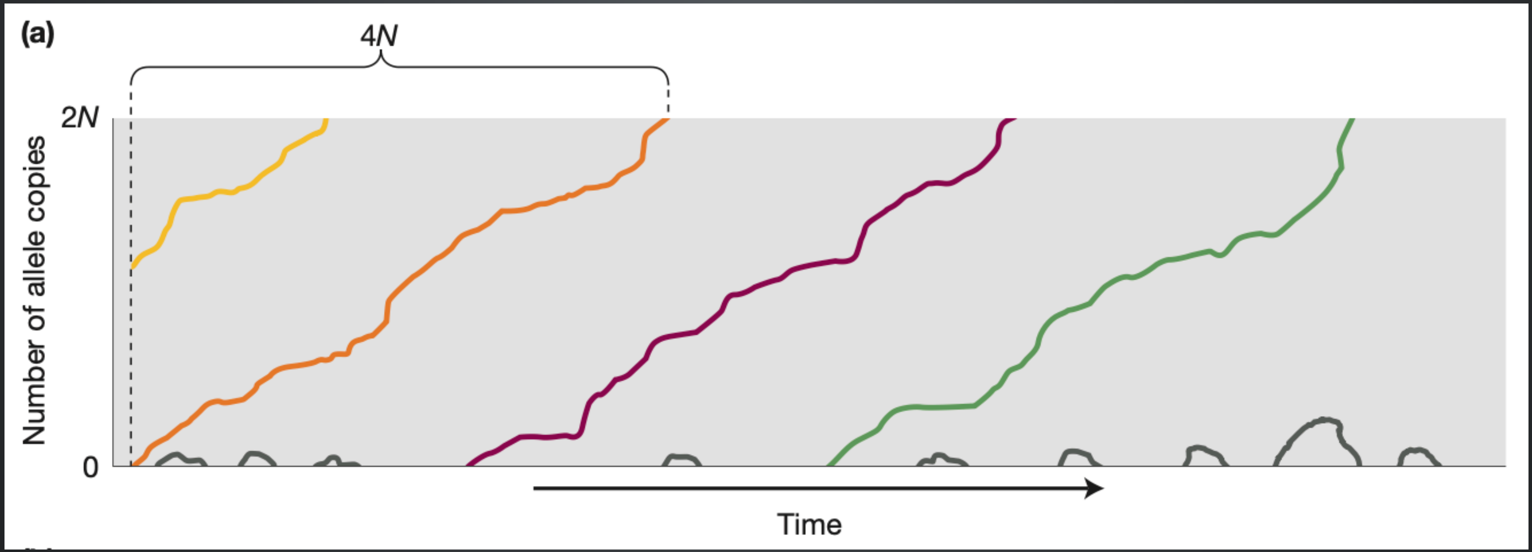

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    - or it may not pass on the beneficial mutation (since by definition, a new mutation must be a single copy in a diploid individual)

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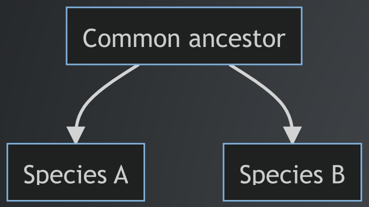
- that is, the substitution rate for neutral mutations equal the rate at which they appear. this is only true for neutral mutations!

MOLECULAR CLOCK: IMPLICATION OF GENETIC DRIFT



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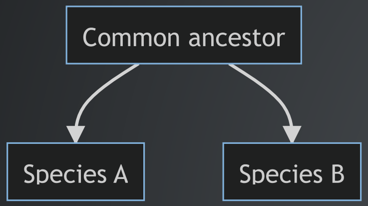

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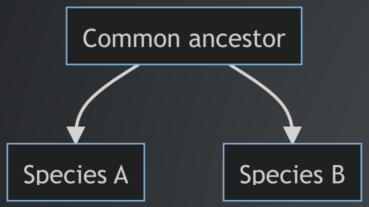

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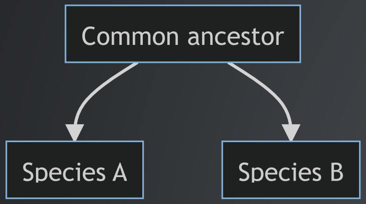

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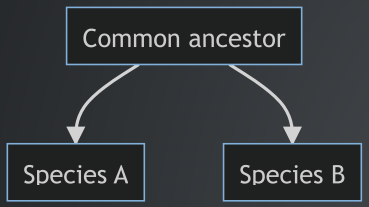

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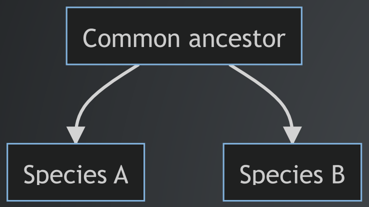

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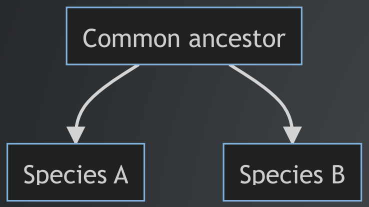

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- can we observe all neutral substitutions?
- how do we know mutation rate?

GENETIC DRIFT AND POPULATION SIZE

Key concept

- Population size is a key factor affecting genetic variation in populations.
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- Most new mutations are lost from populations by drift.
- The constant rate of substitution for neutral mutations can be used to estimate divergence time.

SELECTION

Natural selection (*Darwinian*)

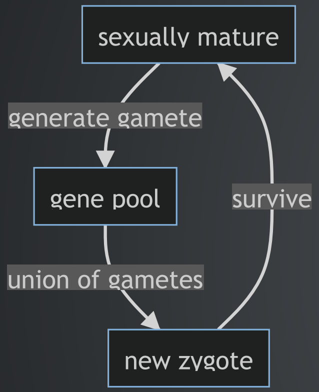
Populations change (*evolve*) over time as the environment (*nature*) favors (*selects*) features that enhance the ability to survive and reproduce.

SELECTION

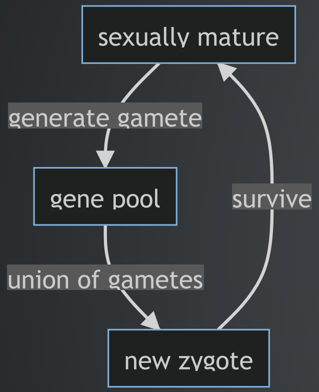
Question

We often use the phrase "*survival of the fittest*" to describe Darwinian selection. If an individual is physically strong, resistant to disease, and lives a long life, does this individual have a higher fitness?

SELECTION

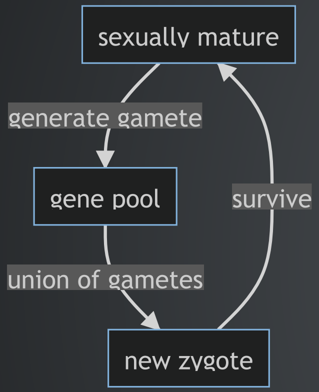

sexually mature
generate gamete
gene pool
survive
union of gametes
new zygote

SELECTION



- **Natural selection** means individuals with certain features are more likely to *survive AND reproduce* than other individuals lacking these features.

SELECTION



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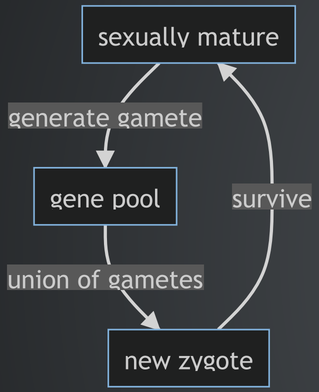

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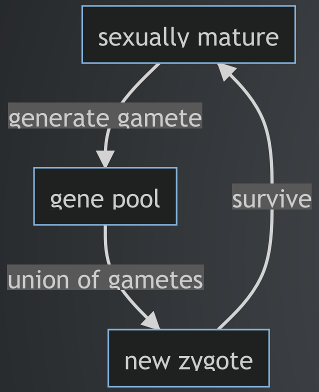

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- Where does selection operate in the left diagram?

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

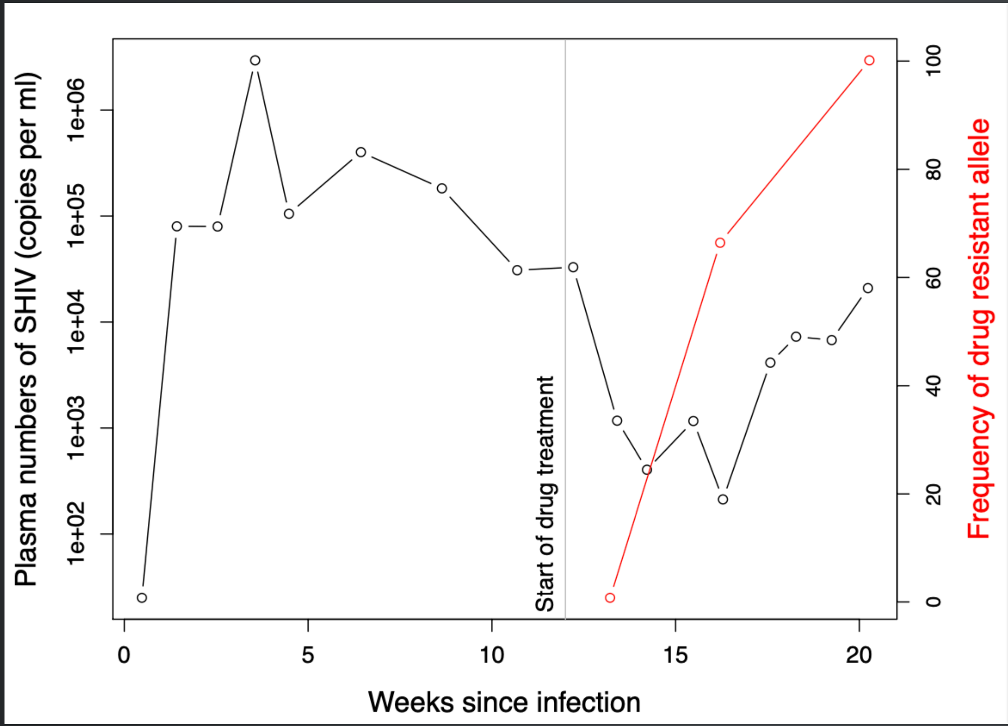

Figure 10.1: The rapid evolution of drug-resistant SHIV. The viral load of SHIV in the blood of a macaque (black line), the frequency of a drug resistance mutation (red line). Data from Feder et al. (2017).

1: Graham Coop, Population and Quantitative Genetics, v1.2, 2020 https://github.com/cooplab/popgen-notes/releases/tag/v1.1

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

- Consider a locus A with three genotypes, A/A, A/a, and a/a in a population. The initial allele frequencies are $p = 0.1, q = 0.9$

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Genotype	**A/A**	**A/a**	**a/a**
Average number of offspring (W)	10	10	5
Relative fitness (w)	1.0	1.0	0.5
Genotype frequency			

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- The relative contribution of each genotype to the gene pool is determined by **the product of its fitness and frequency**

Genotype	**A/A**	**A/a**	**a/a**	**Sum**
Relative contribution	1 x 0.01	1 x 0.18	0.5 x 0.81	
	= 0.01	= 0.18	= 0.405	0.595

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    - $p' = ?$ how much did p change over one round of selection?
    - $P_{AA} = ?$, etc.

HOW NATURAL SELECTION ALTERS ALLELE FREQUENCY?

More generally

define $\bar{w} = p^2 w_{A/A} + 2pq w_{A/a} + q^2 w_{a/a}$ is the mean relative fitness of the population

then, $p' = \frac{p^2 w_{A/A} + \frac{1}{2} \times 2pq w_{A/a}}{\bar{w}} = p \frac{p w_{A/A} + q w_{A/a}}{\bar{w}}$

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we define `$w_A = pw_{A/A} + qw_{A/a}$` as allelic fitness, or mean fitness of allele A

`$\Delta p = p' - p = p \frac{w_A}{\bar{w}} - p = \frac{p(w_A - \bar{w})}{\bar{w}}$`

FORMS OF SELECTION

1. Directional selection

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    - Positive selection

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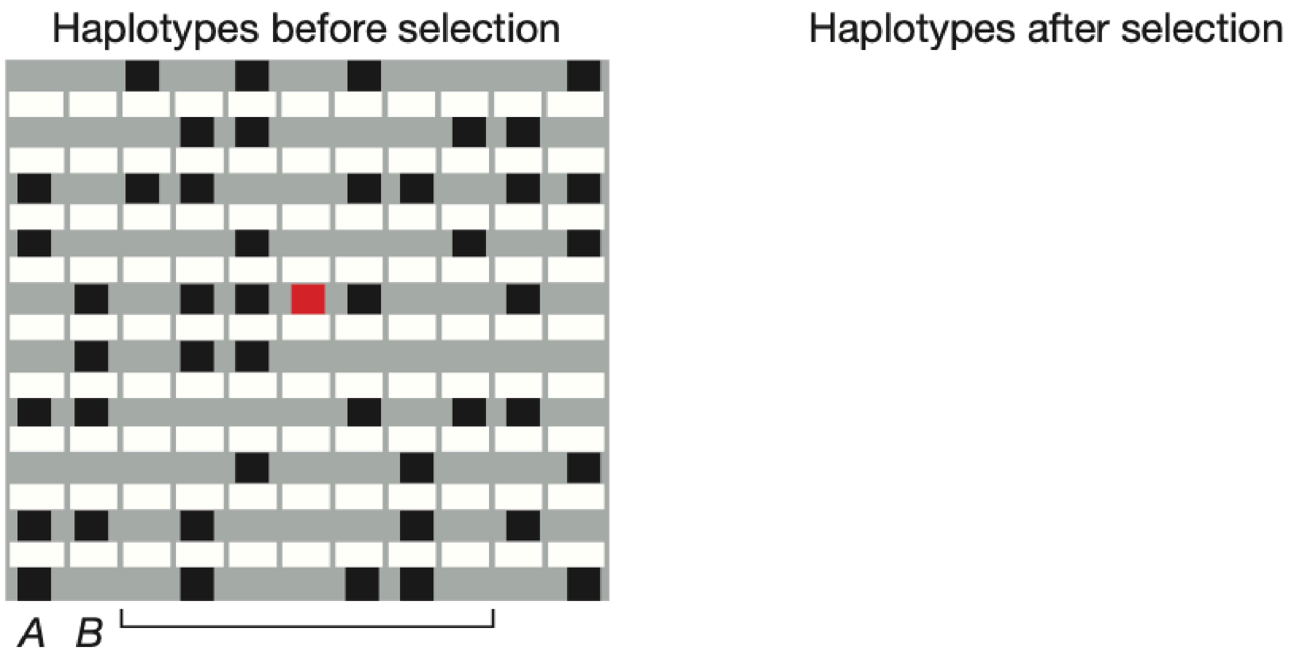
    - can you think of example scenarios for balancing selection?

FORMS OF SELECTION

1. Directional selection
    - Positive selection
    - Purifying (negative) selection
2. Balancing selection
    - if the heterozygous class has higher fitness than either homozygous ones, selection would favor the **maintenance of both alleles**.
    - can you think of example scenarios for balancing selection?
    - are there other types of selection that can **maintain genetic diversity** rather than **eliminating it**?

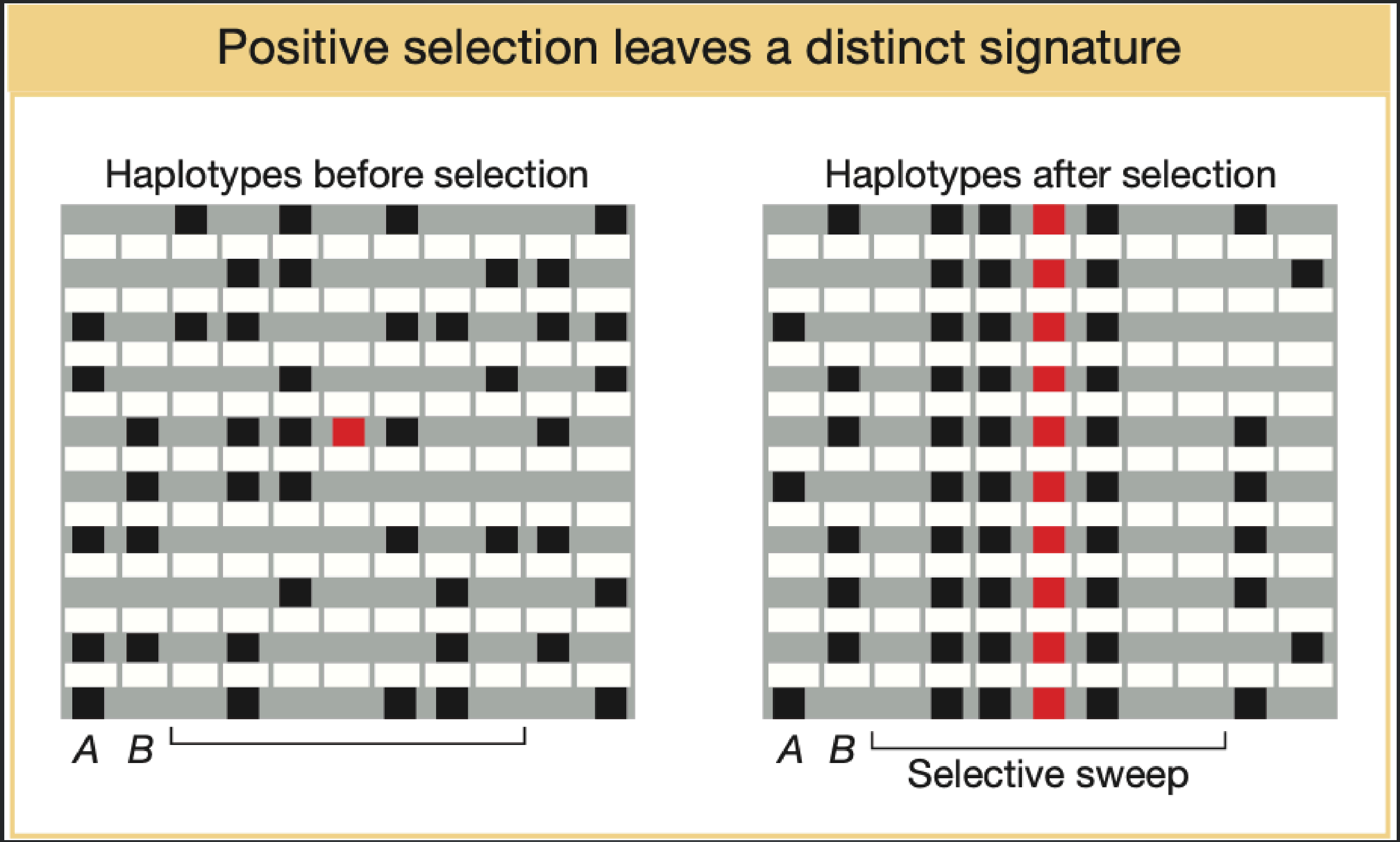
SIGNATURES OF POSITIVE SELECTION

Positive selection leaves a distinct signature



1: Introduction to Genetic Analysis, ed 11, Fig. 18-22

SIGNATURES OF POSITIVE SELECTION



1: Introduction to Genetic Analysis, ed 11, Fig. 18-22

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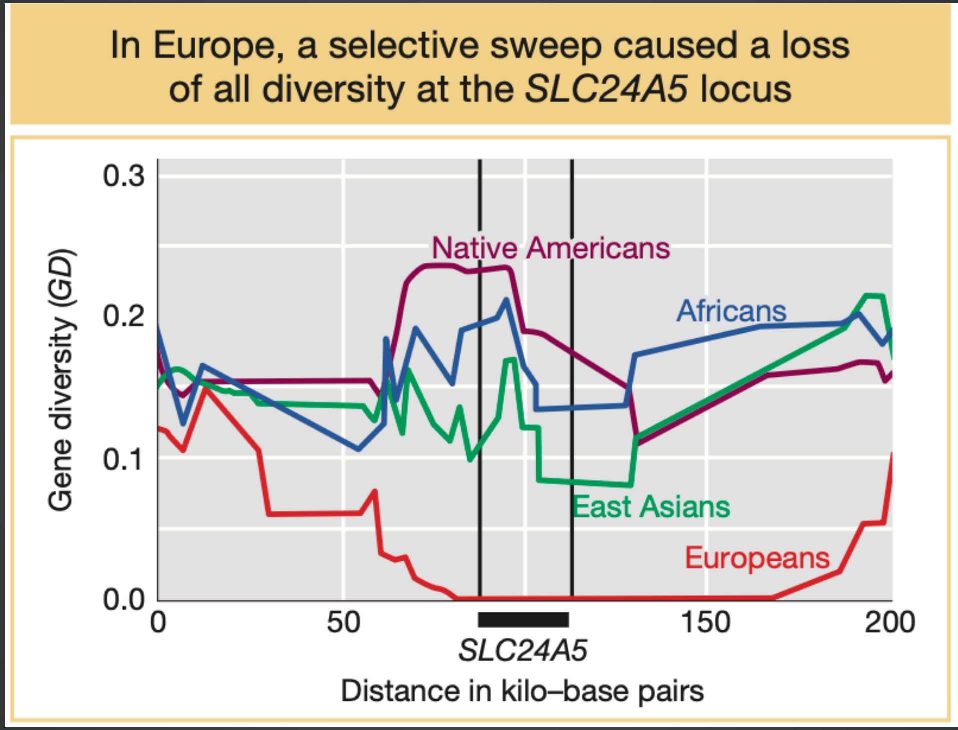
- By driving the favored allele to fixation "very fast", positive selection causes a selective sweep that ends up
    - also fix linked sites or drive the linked allele to high frequency
    - hence removing genetic variation from the population
    - creates LD
- Note that this signature is **transient**. What does this mean, why?

USE SELECTIVE SWEEP SIGNATURE TO INFER POSITIVE SELECTION



1: Introduction to Genetic Analysis, ed 11, Fig. 18-23

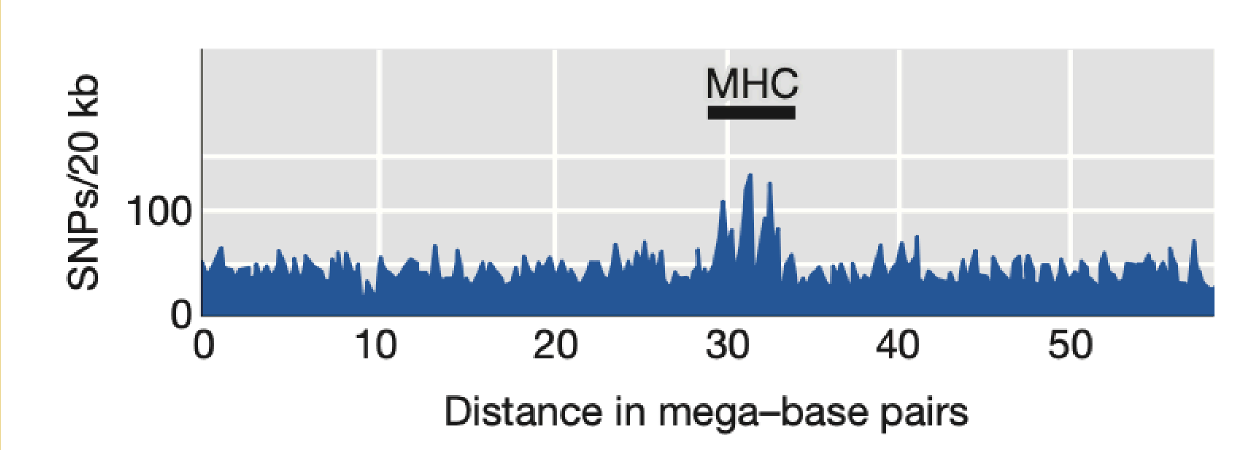
TABLE 8-6 Some Genes Showing Evidence for Natural Selection in Specific Human Populations

Gene	Presumed Trait	Population
EDA2R (ectodysplasin A2 receptor)	Male pattern baldness	Europeans
EDAR (ectodysplasin A receptor)	Hair morphology	East Asians
FY^{null} (Duffy antigen)	Resistance to malaria	Africans
G6PD (glucose-6-phosphate dehydrogenase)	Resistance to malaria	Africans
Hb (hemoglobin B)	Resistance to malaria	Africans
KITLG (KIT ligand)	Skin pigmentation	East Asians and Europeans
LARGE (glycosyltransferase)	Resistance to Lassa fever	Africans
LCT (lactase)	Lactase persistence; ability to digest milk sugar as an adult	Africans, Europeans
LPR (leptin receptor)	Processing of dietary fats	East Asians
MC1R (melanocortin receptor)	Hair and skin pigmentation	East Asians
MHC (major histocompatibility complex)	Infectious disease resistance	Multiple populations
OCA2 (oculocutaneous albinism)	Skin pigmentation and eye color	Europeans
PPARD (peroxisome proliferator-activated receptor delta)	Processing of dietary fats	Europeans
SI (sucrase-isomaltase)	Sucrose metabolism	East Asians
SLC24A5 (solute carrier family 24)	Skin pigmentation	Europeans and West Asians
TYRP1 (tyrosinase-related protein 1)	Skin pigmentation	Europeans

Source: P. C. Sabeti et al., *Science* 312, 2006, 1614–1620; P. C. Sabeti et al., *Nature* 449, 2007, 913–919; B. F. Voight et al., *PLoS Biology* 4, 2006, 446–458; J. K. Pickrell et al., *Genome Research* 19, 2009, 826–837.

1: Introduction to Genetic Analysis, ed 11, Table 18-6

Balancing selection can lead to regions of unusually high genetic diversity



1: Introduction to Genetic Analysis, ed 11, Fig. 18-24